

Part III — The context: fleet, retirements, new engines, and the translation table

Descriptive reference. Sources are cited inline and listed at the end. Takes no view.

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PART III

The context: fleet, retirements, new engines, and the translation table

This Part explains why the CFM56 aftermarket is where it is in 2026: how many engines there are, why far fewer of them retired than their makers planned, what went wrong with the engines meant to replace them, how few new aircraft the airframers delivered, and how lessors and airlines behaved as a result. It ends with a translation table that turns each of those facts into a specific requirement on the aftermarket. A reader who finishes it will be able to name the driver behind any shop-visit, used-part, lease or capital figure in Parts IV to XI.

WHERE THIS PART SITS IN THE BUILD

The primer is built in layers. Parts I and II are first principles: the engine as a machine, and the money that moves around it through leases, reserves and financing. Part III is the context shock. It takes the fleet those first principles describe and asks what happened to it between 2019 and 2026: which engines retired and which did not, why the replacement engines and airframes arrived late, and how owners behaved when they could not get new aircraft. None of that changes how a turbofan works or how a lease is priced. It changes the quantities: how many engines need a shop visit, how many are torn down, how many spares are needed and for how long, and what an extension costs. Section 24 turns each documented driver into a numbered row stating the requirement it creates and what it leaves untouched. Parts IV to XI, the segments, open by naming the rows they answer. This Part depends on Parts I and II for its terms and on nothing written later.

18. The fleet and the installed base

Every shop visit, every used part and every lease described in this primer is sold against an engine that already exists. So the first question is how many engines exist, which state each is in, and how hard it flies. This section defines demand and the states a fleet can be in, then sets out every count of the CFM56 fleet the research reached, with what each count covers. It closes with how mature that fleet is, meaning how many of its engines have never been to a shop.

AFTERMARKET DEMAND

Aftermarket demand for an engine type is the sum of the work and assets needed to keep that type flying: shop visits, modules, spare engines, used serviceable material and leases of engines and aircraft. It is generated by the installed base multiplied by how hard that base flies, net of the engines that leave. Every row of the translation table in §24 names which of these requirements a driver raises. [D9 §2.1, definition assembled in the research]

Part I §2.2 boxed the installed base as the population of engines fitted to aircraft in the operating fleet. It noted that the published "in service" counts may include spare and stored engines, since no source states its basis. That reminder matters here. A spare engine in a warehouse and an engine on a parked aircraft generate no flight cycles and so no maintenance need. The firms that count fleets, Cirium (a data provider) and IBA (an appraiser that also publishes fleet data), sort aircraft into three states.

FLEET STATES: IN SERVICE, PARKED (STORED), RETIRED (WITHDRAWN)

An aircraft in service is assigned to an operator and flying, or in short-term maintenance. A parked or stored aircraft is not flying but has not been formally withdrawn; an aircraft parked for more than about 90 days usually counts as stored, and a stored aircraft can return to service or go on to retirement. A retired or withdrawn aircraft has formally left the operating fleet: it is scrapped whole, parted out so its engines and components can be sold separately (Part V §32), or converted to another use, most commonly a passenger-to-freighter (P2F) conversion (Part II §15). [D9 §2.1]

The states move in both directions except the last. Stored aircraft returned to service in volume after 2021. A retired aircraft does two opposite things to the aftermarket at once. It removes a future customer for shop visits, and it adds supply of used parts and of engines with life remaining. That double effect is why the retirement rate in §19 appears on both sides of the translation table.

AVERAGE AGE

The average age of a fleet is the mean number of years since delivery of its aircraft. Maintenance need rises with age in steps rather than smoothly: the first shop visit, the first life-limited-part replacement, the first heavy airframe check. The retirement decision is largely age-related cost set against what the aircraft is worth kept (§19.1). The average age of the world's commercial aircraft was about 15 years at December 2025, per the airline trade body IATA (the International Air Transport Association) as cited in trade press. [D9 §2.1; IATA Global Outlook, Dec 2025, as cited by Aviation Business News]

18.1 EVERY COUNT OF THE CFM56 FLEET

Counting the CFM56 fleet sounds simple and is not. Part I §2.2 gave the headline counts; this section adds what each count covers and why the basis matters for everything that follows. The research reached thirteen figures from eight sources at three levels: the manufacturer's own pages and investor materials, trade press, and a consultancy blog. None of the "in service" figures states whether spare engines and engines on stored aircraft are included. The research recorded that as an open question. [D9 U3; reconciliation R-001]

EXHIBIT 3.1 — EVERY CFM56 FLEET FIGURE IN THE RESEARCH: WHAT EACH COUNTS, WHETHER SPARES AND STORED ENGINES ARE INCLUDED, SOURCE, LEVEL AND DATE

Figure	What it counts, as the source states it	Spares and stored engines	Source and level	Date
>35,000	CFM56 delivered, all variants, across the 737, A320 and military platforms	Not applicable	CFM website (OEM web page); Aviation Business News (trade press)	Undated; Sept 2025
"well over 30,000"	CFM56 delivered	Not applicable	One dossier's general statement, unsourced (reconciliation R-069)	Not dated
~24,000	CFM56 "in service"	Not stated	Aviation Business News (trade press)	Sept 2025
~23,000	CFM56 "in service"	Not stated	CFM website (OEM web page)	Undated, accessed 2026-10-03
~23,000	CFM56 in service	Not stated	GE Aviation & GECAS Investor Day (18 Jun 2019) and Safran media (2024); which document gave which reading was not captured	2019; 2024
>22,800	CFM56 in Safran's "civil installed base"	Not stated; "civil" excludes military engines	Safran FY2025 results presentation (OEM investor material)	13 Feb 2026 (end-2025 figure)
"over 20,000"	CFM56-5B and -7B in service, the two variants together	Not stated	Chromalloy press release (a parts maker)	30 Oct 2025
>19,000	CFM56 "flying"	"Flying" suggests installed and active; not stated	Safe Fly Aviation (consultancy blog)	2026
~14,200	"Active" CFM56, read by one dossier as all variants (~6,200 -5B and ~7,500 -7B) and by another as the -7B fleet alone	"Active" suggests installed and flying; not stated	Safe Fly Aviation (consultancy blog)	2026

Figure	What it counts, as the source states it	Spares and stored engines	Source and level	Date
"in excess of 8,000"	CFM56-7B "currently installed in 737NGs"	Installed only, by its wording	CFM product page (OEM web page)	Undated
>15,000	CFM56-7B delivered	Not applicable	CFM website; GE Aerospace article on the 15,000th engine	2019 and later
>7,000 aircraft	737NG aircraft delivered with the -7B, two engines each	Not applicable	CFM product page	Undated
>4,100 aircraft	A320ceo-family aircraft delivered with the -5B, two engines each	Not applicable	CFM product page	Undated

Source: CFM International, "The CFM56 engine family" and "CFM56-5B & CFM56-7B" product page, undated, accessed 2026-10-03; Aviation Business News, "CFM56 turbofan aircraft engine", Sept 2025; GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran, FY2025 Results & Investor Update, 13 Feb 2026; Chromalloy, release of 30 Oct 2025; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); GE Aerospace, 2019. Reconciliation R-001, R-002, R-003, R-069; E.1 rules 3 and 14.

Three figures sit at manufacturer level and close together: about 23,000 on CFM's own page, about 24,000 in trade press citing September 2025, and more than 22,800 in Safran's end-2025 count of its civil installed base. [CFM website, undated; Aviation Business News, Sept 2025; Safran, 13 Feb 2026] A fourth figure of about 23,000 was recorded by the shop-visit research from GE's 2019 investor day and from Safran media in 2024, without capturing which document gave which reading. [GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024; D2 §4.2] If both the 2019 and the 2024 readings are about 23,000, the count barely moved in five years. That is consistent with the low retirement rates in §19, but it is not evidence of them, because neither figure states its basis.

The two Safe Fly Aviation figures come from a consultancy blog and are labelled as such wherever they appear in this primer. [E.1 rule 3] The research read the 14,200 figure two ways. One dossier labelled it the CFM56-7B active fleet. Another labelled it the whole active CFM56 fleet, split about 6,200 -5B and 7,500 -7B. [Safe Fly Aviation, 2026; reconciliation R-002] The same publisher's ">19,000 flying" cannot be reconciled with 14,200 from the material reached. The gap of nearly 5,000 engines may be stored engines, spares, or the older -3, -5A and -5C variants, and the primer records both figures without choosing. [reconciliation R-001]

Why the basis matters is arithmetic. The forecast curve in §23 counts about 2,300 CFM56 shop visits a year. Divided into 24,000 engines, that is one visit per engine roughly every ten years. Divided into 14,200, it is one every six. Which base is right decides whether the fleet visits shops more often or less often than Part I's mature interval of 10,000 to 15,000 cycles implies. Worked

example 3.6 does that arithmetic on every base. Here the point is only that the question is live and the sources do not settle it.

18.2 THE SPLIT BETWEEN -5B AND -7B, AND THE V2500 COUNT

The split of the fleet between the two variants that matter for FTAI is unknown. Only the consultancy blog gives one, about 6,200 -5B and 7,500 -7B, and the research read even that figure two ways. [Safe Fly Aviation, 2026; reconciliation R-002] The manufacturer's product page gives delivery counts by airframe instead: more than 7,000 737NGs delivered with the -7B and more than 4,100 A320ceo-family aircraft delivered with the -5B, each aircraft carrying two engines. [CFM product page, undated] Two engines per aircraft gives more than 14,000 -7Bs and more than 8,200 -5Bs installed at delivery, about 22,200 together before any retirement or any spare engine is counted. That is arithmetic on the manufacturer's own figures, done for this Part. It is a check, not a count. It sits close to the "in service" totals above only if few engines have retired and if those totals include spares, and the research cannot say whether either condition holds.

The same page says "in excess of 8,000 CFM56-7Bs are currently installed in 737NGs", with no date visible. [CFM product page, undated; reconciliation R-003] At two engines per aircraft that is 4,000 737NGs, well under the 7,000 delivered. The research recorded that the statement may be old, from early in the programme, or may count a subset, and did not resolve it. Part I §2.2 carries the same tension. [D9 T5]

The V2500 count is the second unknown. Part I §10.1 boxed the V2500, the A320ceo's alternative engine from the IAE consortium. Aviation Week's 2023 fleet forecast put 5,286 V2500s in service in 2023, of which 5,260 were the -A5 and 26 the older -A1. [Aviation Week, V2500 data tool / 2023 Fleet & MRO Forecast] A market-research page gives ">13,000 units" in service in mid-2026; the research flagged the page as low grade. [MarketIntel, 2026; reconciliation R-010] The research's own check runs as follows. The -5B powered more than 4,100 of the roughly 8,000 or more A320ceos delivered, leaving on the order of 4,000 V2500-powered A320ceos plus the MD-90s, so roughly 8,000 installed V2500s at most before retirements. [D9 T8, arithmetic on the sources' figures] The primer carries 5,286 as the sourced count and the ">13,000" only as a low-grade page. A current count from IAE or from RTX (Pratt & Whitney's parent) was not reached. [D9 U4]

EXHIBIT 3.2 — THE AIRFRAMES AND THE OTHER ENGINES: FLEET-LEVEL FIGURES REACHED, WITH SOURCE AND DATE

Item	Figure	Source and level	Date
A320 family (ceo and neo together): orders / deliveries / in service	20,378 / 12,798 / 11,308	Wikipedia order-and-delivery list (secondary compilation)	July 2026

Item	Figure	Source and level	Date
A320ceo-family aircraft stored 90+ days, and their average age	~570; 17.1 years	Naveo, AFRA (an aircraft-recycling trade association) panel (consultancy presentation)	6 Jun 2023
Share of 737NGs / A320ceos aged ten years or less	47% / 45%	ISTAT (the International Society of Transport Aircraft Trading) via ePlaneAI	Undated; probably 2021–22 (D9 U2)
Average age of the commercial fleet	~15 years	IATA Global Outlook via Aviation Business News	Dec 2025
LEAP (CFM's new-generation engine) in service	~5,500; ~50% under long-term service agreements against ~15% of the CFM56 fleet	Aviation Week (trade press)	Undated
V2500 in service	5,286 (5,260 -A5; 26 -A1)	Aviation Week 2023 Fleet & MRO Forecast	2023
V2500 "in-service fleet"	">13,000 units"	MarketIntelto (market-research page; low grade)	Mid-2026
737-800 freighters in service / conversion pipeline	>260 / >300	Acumen Aviation; Cargo Facts (trade press)	Mid-2025
737-700 freighter conversions to date	<50	Acumen Aviation	18 Dec 2025
Age at narrowbody freighter conversion (737, 757)	21–22 years on average	Cargo Facts	Date not captured
737 MAX (the re-engined 737) cumulative deliveries	2,399	Wikipedia order-and-delivery list	2026

Source: Wikipedia, "List of Airbus A320 family orders and deliveries", July 2026, and "List of Boeing 737 MAX orders and deliveries", 2026; Naveo Consultancy, AFRA Aircraft & Engine Panel presentation, 6 Jun 2023; ISTAT, "CFM International CFM56-5B/-7B", undated; Aviation Business News, "CFM56 turbofan aircraft engine"; Aviation Week, "Leap Aftermarket Poised For Growth", undated, and "Data Tool: The Future Of IAE V2500 Engine"; MarketIntelto, "V2500 MRO Market Research Report 2034"; Acumen Aviation, 737-700 market insight, 18 Dec 2025; Cargo Facts, two articles, dates not captured. D9 §4.1; reconciliation R-010.

Two things in that table are themselves unknowns. The combined A320 figure of 11,308 in service is not split between ceo and neo, so the number of A320ceos flying in 2026 was not reached. [D9 U1] The "47% / 45% aged ten years or less" statistic is undated; since -7B production

ended in 2019 and -5B production in 2022, the share must be lower in 2026, and the research judged the figure to be from about 2021 or 2022. [D9 U2]

18.3 HOW MATURE THE FLEET IS

A second way to describe a fleet is by how many shop visits its engines have had. The research used the phrase "fleet maturity" for this without defining it; here it means the share of the installed base by the number of shop visits endured, none, one, or two or more. It matters because an engine that has not yet had a first visit is a visit still to come, and because the first visit on a mature engine is usually the heaviest event of its life so far. Part I §6.2 explained that first runs are long, with the latest -5B and -7B variants often flown almost to their life-limited-part limit before the first removal.

EXHIBIT 3.3 — CFM56 FLEET MATURITY: SHARE OF ENGINES BY SHOP VISITS ENDURED, THREE READINGS

Reading	No shop visit yet	One shop visit	Two or more	Source and level	Date
GE investor day	57%	21%	22% (by subtraction)	GE Aviation & GECAS Investor Day (OEM investor material)	18 Jun 2019
Safran media	About 70% with zero or one, not split		About 30% (by subtraction)	Safran media (OEM)	2024
Consultancy blog	"over half"	"another quarter"	The remainder	Safe Fly Aviation (consultancy blog)	2026

Source: GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024, via Aviation Week; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog). Compiled in D2 §4.2; reconciliation R-012.

The three readings are not on one base and are not all figures. The two manufacturer readings put the zero-or-one share at 78% in 2019 and about 70% in 2024. The consultancy blog's "over half" plus "another quarter" is about 75% in 2026, which would be higher than Safran's 2024 share. [reconciliation R-012] No engine loses a shop visit, so a rising share can only come from retirements removing multi-visit engines from the denominator, or from the readings being on different bases. The primer records the three readings and stops.

WORKED EXAMPLE 3.1 — HOW MANY CFM56 ENGINES HAVE NEVER BEEN TO A SHOP, ON EACH BASE AND EACH READING

Inputs. Shares (sourced, Exhibit 3.3): 57% with no first visit and 21% with one (GE, June 2019); about 70% with zero or one (Safran, 2024); "over half" with none and "another quarter" with one (Safe Fly Aviation, consultancy blog, 2026). Bases (sourced, Exhibit 3.1): about 23,000 (CFM,

undated; also the 2019 and 2024 readings via D2); about 24,000 (Aviation Business News, Sept 2025); more than 19,000 and about 14,200 (Safe Fly Aviation, consultancy blog, 2026). Each share is applied to the base from its own source or date; no share is applied to another publisher's base without saying so.

Step 1, the 2019 reading on the 2019 base.

no first visit: $0.57 \times 23,000 = 13,110$ engines
 one visit: $0.21 \times 23,000 = 4,830$ engines
 two or more: $23,000 - 13,110 - 4,830 = 5,060$ engines

Step 2, the 2024 reading. Safran did not split zero from one.

zero or one visit, on 23,000: $0.70 \times 23,000 = 16,100$ engines
 zero or one visit, on 24,000: $0.70 \times 24,000 = 16,800$ engines

Step 3, the 2026 consultancy reading on the consultancy's own bases. "Over half" is taken as at least 50% and "another quarter" as 25%, which turns words into floor figures.

on >19,000: no visit $\geq 9,500$; one visit $\approx 4,750$
 on ~14,200: no visit $\geq 7,100$; one visit $\approx 3,550$

Step 4, what the readings say together. The zero-or-one share moved from 78% to about 70% between the two manufacturer readings. On a 23,000 base that is about 1,840 engines. It is not a count of visits performed, because retirements of older, multi-visit engines also move the share, in the other direction. The consultancy's 2026 share of about 75% sits above the 2024 share and cannot be reconciled with it on one base (reconciliation R-012).

So on the manufacturer readings, somewhere between 13,000 and 17,000 CFM56 engines had seen at most one shop visit in 2019 to 2024, which is the pool from which the first-shop-visit peak in 2027 (§23.1) is drawn. Limits of the example: every share is on an unstated base; the dates differ by up to seven years; two of the shares are words rather than figures; and the -5B/-7B split of the pool is unknown (§18.2). [GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024; Safe Fly Aviation, 2026; D2 §4.2; reconciliation R-001, R-012]

19. Retirements: how the decision is made and what happened 2019–2026

Part I §2.3 boxed the retirement rate as the share of a fleet withdrawn in a year, and gave GE's falling planning assumption in one sentence. This section is the full treatment. It explains how an owner decides to retire an aircraft, sets out what actually retired between 2019 and 2025 with the two data providers' conflicting counts, follows GE's planning number down from 3–4% to

"below 1%", and works the arithmetic of what one and three-quarter points of retirement rate mean in engines a year.

19.1 HOW THE DECISION IS MADE

An aircraft is retired by its owner, an airline or a lessor, when keeping it in service costs more than it is worth kept, and when the alternatives of selling it on, converting it to a freighter or parting it out are better. Five inputs enter that decision. [D9 §2.3]

The first input is age, and the way maintenance cost arrives in lumps rather than smoothly. An engine performance-restoration shop visit opens the core to restore exhaust-gas-temperature margin (Part I §5 and §6). A life-limited-part (LLP) replacement forces the engine open when a rotating part reaches its certified cycle limit (Part I §4). A heavy airframe check does the same to the airframe. Each is a multi-million-dollar event. Acumen Aviation's December 2025 note on the 737-700 states that "as the 737-700 fleet ages, heavy checks and major engine shop visits increasingly exceed the residual value of the aircraft, driving earlier retirements or conversions".

[Acumen Aviation, 18 Dec 2025] A 2023 analysis found that aircraft become vulnerable to retirement at 16 to 18 or more years, particularly if the owner does not want to pay for the next heavy check or shop visit, and that the roughly 570 A320ceo-family aircraft then stored for 90 or more days had an average age of 17.1 years. [Naveo, AFRA panel, 6 Jun 2023]

MAINTENANCE CLIFF

The maintenance cliff is the point at which the next heavy maintenance event, an engine shop visit, a life-limited-part replacement or a heavy airframe check, costs more than the aircraft is worth kept. An owner at the cliff retires, converts or sells the aircraft rather than pay for the event. Aircraft become vulnerable to it at about 16 to 18 years and older. Aviation Week's rule of thumb that a CFM56 needs a shop visit roughly every eight years makes 24 years a de facto retirement date, because a third full overhaul is "deemed too expensive". [D9 §2.3; Naveo, 6 Jun 2023; Aviation Week, "Retirement Uptick Will Restock Used Engine Parts", date not captured]

The second input is lease economics. At the end of an operating lease (Part II §12) the lessor chooses between extending with the current airline, moving the aircraft to a new lessee, or selling or parting it out. The airline, for its part, chooses between extending and returning. Both choices turn on whether a replacement aircraft is available, and between 2023 and 2026 it usually was not (§22).

LEASE TRANSITION

A lease transition is the move of an aircraft from one lessee to the next. Between the two leases the lessor pays for reconfiguration of the cabin, for the maintenance needed to meet the next lessee's delivery condition, and for the downtime while the aircraft earns no rent. Transition is the alternative to extension (§22.3), and its cost is one reason a lessor extends when it can. [D9 §2.3]

The third input is freighter conversion, which Part II §15 owns. A passenger-to-freighter conversion gives a 20-plus-year-old airframe a second life. Cargo Facts reports narrowbody freighters, the 737 and 757, being converted at 21 to 22 years of age on average. [Cargo Facts, date not captured] More than 260 737-800 freighters were in service as of mid-2025 with a conversion pipeline exceeding 300, against fewer than 50 737-700 conversions. [Acumen Aviation, 18 Dec 2025; Cargo Facts] A conversion candidate still needs engines with life on them. So conversion is a source of demand for engines with green time (the remaining usable life on an engine before its next mandatory shop visit, Part II §16) or for shop visits, and not a source of teardown supply.

The fourth input is part-out value (Part V §32). When the value of an aircraft as a sum of parts exceeds its value flying, the owner parts it out. For a mid-life CFM56-powered narrowbody the engines are the dominant share of that sum, and Part V §32 works the numbers. The consultancy figures the research reached are that a run-out CFM56 is bought as feedstock for \$0.8–1.2M, yields \$1.6–2.2M of used material, and nets \$0.4–0.8M before repair cost. [Safe Fly Aviation, 2026, consultancy blog] Rising used-part prices raise part-out value and so, at the margin, encourage retirement. That is a stabilising feedback. The research records that it was not strong enough to lift retirements in 2024 to 2026. [D9 §2.3]

The fifth input is replacement availability, and it governs the other four.

REPLACEMENT AVAILABILITY IS THE GATE

None of the first four inputs triggers a retirement if the airline cannot get a replacement aircraft. An airline at the maintenance cliff with no new aircraft to fly pays for the shop visit anyway, because the alternative is to shrink. A lessor whose lessee cannot find a replacement extends the lease rather than take the aircraft back into a market with no buyer-operators. This is the single mechanism through which the new-engine and airframer problems of §20 and §21 act on the old fleet: they close the gate. Cirium Ascend's count that only about a dozen mid-life 737NGs and fewer than 25 A320neos were available to rent anywhere in the world is the measure of how closed it was. [D9 §2.3; Cirium Ascend via Bloomberg, 2023–24 context]

EXHIBIT 3.4 — INPUTS TO THE RETIREMENT DECISION AND THE FIGURE THE RESEARCH REACHED FOR EACH

Input	Figure reached	Source and level	Date
Age at which aircraft become vulnerable to retirement	16–18+ years	Naveo, AFRA panel (consultancy presentation)	6 Jun 2023
Stored A320ceo-family aircraft (90+ days) and their average age	~570; 17.1 years	Same	6 Jun 2023
Shop-visit rule of thumb and the de facto retirement age it implies	About every 8 years; 24 years	Aviation Week (trade press)	Date not captured
737NG delivery rate in 1998–2002, the cohort now at the cliff	~230 a year	Aviation Week	Date not captured
Age at freighter conversion, 737 and 757	21–22 years on average	Cargo Facts (trade press)	Date not captured
737-800 freighters in service / pipeline	>260 / >300	Acumen Aviation; Cargo Facts	Mid-2025
737-700 freighter conversions to date	<50	Acumen Aviation	18 Dec 2025
Run-out CFM56 as feedstock: purchase / used-material yield / net	\$0.8–1.2M / \$1.6–2.2M / \$0.4–0.8M	Safe Fly Aviation (consultancy blog)	2026
CFM56-7B half-life market value, the "kept flying" side of the comparison	\$5.7M (-7B24); \$6.4M (-7B27)	IBA (appraiser)	Sept 2025
Mid-life aircraft available to rent worldwide (replacement availability)	~12 737NGs; <25 A320ceos	Cirium Ascend via Bloomberg (data provider via press)	2023–24 context

Source: Naveo Consultancy, AFRA panel, 6 Jun 2023; Aviation Week, "Retirement Uptick Will Restock Used Engine Parts", date not captured; Cargo Facts, "Passenger-to-freighter conversions are taking place on increasingly older aircraft" and "In-service 737NG freighter fleet surpasses 100"; Acumen Aviation, 18 Dec 2025; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); IBA, Engine Values Release 2025B, Sept 2025; Bloomberg via AJOT (the American Journal of Transportation), "Used jets rents surge 44%". D9 §2.3; D3 §2.6 and §2.8.

19.2 AIRCRAFT RETIREMENTS, 2019 TO 2025

The research reached aircraft-level counts from two data providers, one appraiser and one consultancy, and an engine-level rate from GE. They agree on direction and disagree on level. [reconciliation B-6, R-005] Exhibit 3.5 sets out every figure with the denominator its source states, because the figures are not on one basis and the reconciliation instructs that they never be mixed.

[registry C-10]

EXHIBIT 3.5 — REALISED RETIREMENTS 2019–2025, EVERY FIGURE WITH ITS DENOMINATOR, SOURCE AND DATE

Period	Figure	Denominator, as stated	Source and level	Date
2010–2019 average	Just under 540 a year; peak 694 in 2013	Aircraft part-outs, all types	Cirium Fleets Analyzer via Aerodynamic Advisory at an IATA maintenance conference (data provider via consultancy)	2023
2019	502	Aircraft retirements	Cirium via Aerodynamic Advisory	2023
2020	465	Same	Same	2023
2021	333	Same	Same	2023
2022	325	Same	Same	2023
2023	"Similar" to 2024's fewer than 200	Aircraft; scope not visible	IBA via AviTrader (appraiser via trade press)	24 Sep 2024
2024	Fewer than 200; 70% below the 2020 peak	Aircraft; scope not visible	IBA via AviTrader	24 Sep 2024
2024	587	Aircraft; scope not visible	Cirium Fleets Analyzer via an Informa forecast page	Undated forecast page
2019 → 2024	2.2% → 0.4%	Passenger aircraft retired as a share of the active-plus-stored fleet	Aviation Week / Bain material; exact document not visible	Undated
2024–2026	About 24% below the 2010–2019 norm	Aircraft, all types	McKinsey (consultancy)	Date not captured (2024–25)
Next five years	1,190 A320ceo-family and 795 737NG aircraft, 35% of all retirements	Aircraft	Cirium / Ascend by Cirium	Date not captured
By 2040	About 72% of all 737NGs retired	737NG aircraft	Ascend by Cirium	Date not captured

Period	Figure	Denominator, as stated	Source and level	Date
2024 and 2025	About 1.5% of the installed base in 2025, "broadly consistent with 2024"	CFM56 engines	GE Q4 2025 results via Leeham News	22 Jan 2026
1Q 2026	"Below 1%"	CFM56 engines	GE Aerospace investor-relations page	2026
2025	46.1%	Global Replacement Ratio, 2015–2024 basis	Cirium via Informa	Undated

Source: IATA Maintenance Cost Conference, Aerodynamic Advisory (Murby), "Retirements, part-outs, dismantling", 2023; AviTrader, "Global airline industry set for record growth in 2024" (IBA data), 24 Sep 2024; Informa Markets, Commercial Forecast story, page 5 (Cirium figures); Aviation Week, "Pockets Of Demand Disrupt Expected Narrowbody Retirement Surge"; McKinsey, "What does the future hold for commercial-aviation maintenance?"; Leeham News, 22 Jan 2026; GE Aerospace, "Recent events: your questions answered", 2026. D9 §2.4 and §4.2; D3 §2.6 and §4.4; reconciliation R-004, R-005; E.1 rule 13.

The 2024 count is the tension. IBA, an appraiser, said in September 2024 that aircraft retirements would stay at fewer than 200 for the year, similar to 2023 and 70% below the 2020 peak. [IBA via AviTrader, 24 Sep 2024] Cirium's Fleets Analyzer, as cited on a forecast page, recorded 587 retirements in 2024. [Cirium via Informa] The two cannot both count the same thing. The scope is visible in neither: one may count passenger jets only, one may be a year-to-date figure at the time of writing, one may apply a size threshold. [D9 T2; reconciliation R-005] The series also disagree on where the peak sits. IBA calls 2020 the peak; in Cirium's series 2019, at 502, stands above 2020, at 465. The reconciliation added two further figures from arithmetic on the sources' own numbers, derived in the research and not counts. McKinsey's "24% below the 2010–2019 norm" applied to the just-under-540 average would be about 410 a year. Cirium's projection that 1,190 A320ceo-family and 795 737NG aircraft will be 35% of retirements over five years implies about 5,670 retirements in total, about 1,130 a year. [reconciliation R-005] So the research holds four numbers for roughly the same quantity, fewer than 200, about 410, 587 and about 1,130, with the scope visible in none. The primer presents them and stops. Year-by-year counts for 2023, 2024 and 2025 from one provider on one basis were not reached. [D3 U16]

The passenger-fleet share tells the same story in a different unit. Retirements as a share of the active-plus-stored passenger fleet fell from 2.2% in 2019 to 0.4% in 2024. [Aviation Week / Bain material, undated] Cirium's Global Replacement Ratio expresses the shortfall from the delivery side.

GLOBAL REPLACEMENT RATIO

The Global Replacement Ratio is the number of deliveries needed to replace retirements, expressed as a share of total deliveries over a period. Cirium put it at 46.1% for 2025 on 2015–2024 data: fewer than half of new deliveries were replacing a retired aircraft, and the rest were growth. A low ratio means the fleet is growing by addition rather than substitution, which is the delivery-side face of low retirements. [Cirium via Informa, undated; D9 §2.4]

Named programmes show the decision being made both ways at once. United, WestJet and Copa have started retiring 737NGs in favour of the 737 MAX 8. [Acumen Aviation, 18 Dec 2025] Wizz Air announced a gradual phase-out of its A321ceo fleet on 24 March 2026. [Wizz Air release, 24 Mar 2026] Against those, Lauda Europe, part of the Ryanair group, is extending A320 leases through the early 2030s because the transition to new aircraft proved too costly, and Lufthansa and Ryanair were reported holding onto older aircraft to fill the gap. [Travel Extra, 2026; Bloomberg via AJOT] IBA noted at the same time that delivery delays had led operators to maximise daily utilisation across all age groups, with mature passenger-narrowbody utilisation "rising significantly". [IBA via AviTrader, 24 Sep 2024] That utilisation effect has its own row in the translation table, because it adds cycles without adding engines.

19.3 THE CFM56 ENGINE RATE: GE'S PATH FROM 3–4% TO BELOW 1%

The figure that the rest of the market plans against is GE's. As one of the two partners in CFM, GE sets a retirement assumption for the CFM56 fleet each year and reports against it. The research captured that assumption at four dates in two venues, and the readings do not quite agree with each other.

EXHIBIT 3.6 — GE'S CFM56 RETIREMENT STATEMENTS BY DATE AND VENUE, WITH THE OTHER DENOMINATORS IN USE

Statement	Figure	Denominator	Venue and date
Original planning assumption for 2026 ("last year")	3–4%	CFM56 installed base	GE CFO Rahul Ghai via Aviation Week, 2026 (exact date not visible)
Revised at the start of 2026	2–3%	Same	Same
"Currently seeing something between"	1.5–2%	Same	Same
2025 actual	About 1.5%, broadly consistent with 2024	CFM56 installed base	GE Q4 2025 results via Leeham News, 22 Jan 2026

Statement	Figure	Denominator	Venue and date
2026 expected	About 2%, "below the 2–3% range previously anticipated"	Same	Same
2026 outlook assumption	About 2% of total fleet	CFM56 fleet	GE Aerospace investor-relations page, 2026
1Q 2026 actual	"Below 1%"	Same	Same
Consultancy reading	"Near 2% per year"	About 14,200 "active" CFM56 (consultancy base)	Safe Fly Aviation, 2026 (consultancy blog)
"Normal" rate expected from 2028	About 2.7% a year	All commercial aircraft	McKinsey, 2024–25
Passenger fleet, 2024	0.4%	Passenger aircraft, active plus stored	Aviation Week / Bain material, undated
V2500	"No retirement surge expected over next five years"	V2500 engines	IAE via Aviation Week, 2024/25

Source: Aviation Week, "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; Leeham News, "GE Aerospace FY and Q4 2025 Earnings", 22 Jan 2026; GE Aerospace Investor Relations, "Recent events: your questions answered", 2026; Safe Fly Aviation, 2026 (consultancy blog); McKinsey, date not captured; Aviation Week, "Pockets Of Demand Disrupt Expected Narrowbody Retirement Surge"; Aviation Week V2500 coverage via D2 §4.2. Reconciliation R-004; D9 T3; registry C-10; E.1 rule 13.

The path is the same in both venues: the assumption fell, and the outturn fell further. The readings differ in detail. Aviation Week's quotation of the chief financial officer has the 2026 planning number starting at 3–4% "last year", cut to 2–3% at the beginning of the year, and running at 1.5–2% as "currently seeing". [GE CFO via Aviation Week, 2026] Leeham's reading of the fourth-quarter 2025 call has 2025 at about 1.5%, broadly the same as 2024, and 2026 expected at about 2%, "below the 2–3% range previously anticipated". [GE Q4 2025 results via Leeham, 22 Jan 2026] The two differ by half a point on the 2026 expectation, 2% against 1.5–2%, and on whether 3–4% was ever the 2026 number or an earlier year's. [D9 T3] GE's own investor-relations page then reported first-quarter 2026 retirements "below 1%", against the about 2% it assumed for the year. [GE Aerospace investor-relations page, 2026; reconciliation R-004] A single quarter is not a year, and the page does not say whether the quarterly figure is annualised. The primer records all three readings.

Every other rate in the exhibit is on a different denominator and must not be set beside GE's as if it were the same quantity. [registry C-10] GE counts CFM56 engines. McKinsey's 2.7% counts all commercial aircraft and is a forecast of a return to normal from 2028, not an outturn. [McKinsey, 2024–25] The 0.4% counts passenger aircraft against an active-plus-stored base, which is a larger

denominator than the in-service fleet. The consultancy blog's "near 2%" is stated against its own 14,200 "active" base, which §18.1 showed is itself contested. [Safe Fly Aviation, 2026; reconciliation R-002] For the V2500, IAE expected "no retirement surge" over five years, which is a statement of direction without a rate. [IAE via Aviation Week, 2024/25]

WORKED EXAMPLE 3.2 — WHAT A 1.75-POINT DIFFERENCE IN THE RETIREMENT RATE MEANS IN ENGINES A YEAR

Inputs. Installed base (sourced, Exhibit 3.1): about 24,000 (Aviation Business News, Sept 2025); about 23,000 (CFM website, undated); more than 22,800 (Safran, end-2025). Consultancy base (sourced, labelled): about 14,200 "active" (Safe Fly Aviation, 2026). Rates (sourced, Exhibit 3.6): 3.5%, the midpoint of GE's original 3–4%; 2.5%, the midpoint of 2–3%; 1.75%, the midpoint of 1.5–2%; 1.5%, the 2025 outturn; 2%, the 2026 expectation; below 1%, the first quarter of 2026. The midpoints are this Part's choice for arithmetic and are not GE figures. Every base may include spare and stored engines (D9 U3).

Step 1, engines a year on the 24,000 base.

$$\begin{array}{ll} 3.5\%: 24,000 \times 0.035 = 840 & 2.5\%: 24,000 \times 0.025 = 600 \\ 1.75\%: 24,000 \times 0.0175 = 420 & 1.5\%: 24,000 \times 0.015 = 360 \\ 2.0\%: 24,000 \times 0.020 = 480 & \text{below } 1\%: \text{fewer than } 24,000 \times 0.010 = 240 \end{array}$$

Step 2, the same on the other manufacturer-level bases.

$$\begin{array}{l} \text{on } 23,000: 3.5\% = 805; 1.75\% = 403; 1.5\% = 345; 2.0\% = 460 \\ \text{on } 22,800: 3.5\% = 798; 1.75\% = 399; 1.5\% = 342; 2.0\% = 456 \end{array}$$

Step 3, the 1.75-point gap between the original plan and the realised range.

$$\begin{array}{l} \text{on } 24,000: 24,000 \times (0.035 - 0.0175) = 420 \text{ engines a year} \\ \text{on } 22,800: 22,800 \times 0.0175 = 399 \text{ engines a year} \\ \text{over three years 2024–26: about 1,260 (on 24,000) or about 1,200 (on } 22,800) \end{array}$$

The three-year figures are upper bounds, because the base shrinks a little each year as engines leave (D9 §2.2). Step 4, the consultancy base, shown because Part IX's feedstock arithmetic was first done on it.

$$\text{on } 14,200: 3.5\% = 497; 1.75\% = 249; 2.0\% = 284; \text{ the 1.75-point gap} = 249 \text{ engines a year}$$

Step 5, what 420 engines a year are in the aftermarket's units. In aircraft, 420 engines are 210 narrowbody-equivalents at two engines each. In shop visits, each retained engine eventually needs the visit it would otherwise have avoided; if every one needed one restoration during its retained period, 420 is at most about 18% of a 2,300-visit year ($420 \div 2,300 = 0.183$), spread over several years rather than falling in one, so the annual effect is smaller than 18% and the figure is a ceiling. In

used material, 420 fewer teardowns a year at the consultancy's yield of \$1.6–2.2M of used parts per run-out engine is \$0.67–0.92B a year of used parts not produced (illustrative; consultancy input). In FTAI Power's unit, 100 Mod-1 cores a year (Part IX §74) would be 24% of 420 retirements, 28% of 360, 21% of 480, and 35% of the 284 implied on the consultancy base at 2%, which is the pair of readings the reconciliation requires Part IX to show (reconciliation C.8).

So the gap between what the manufacturer planned and what happened is about 400 engines a year on any manufacturer-level base, or about 1,200 engines over 2024 to 2026, that are still flying and will need work, and that were not torn down for parts. Limits of the example: the base is undefined as to spares and stored engines; the rates come from different dates and venues (D9 T3); the midpoints are the Part's, not GE's; the consultancy base is not a manufacturer figure; and the first-quarter 2026 rate may not hold for the year. [GE CFO via Aviation Week, 2026; GE Q4 2025 results via Leeham, 22 Jan 2026; GE Aerospace investor-relations page, 2026; Aviation Business News, Sept 2025; Safran, 13 Feb 2026; Safe Fly Aviation, 2026; D9 §2.2; reconciliation R-001, R-004, C.8]

19.4 BOTH SIDES OF THE LEDGER

The retirement rate is one number with two effects, and the translation table carries them as separate rows. Lower retirements mean more engines accruing cycles, so more shop visits, more life-limited-part replacements as retained engines reach their limits, and more full restorations in place of green-time run-outs; that is row T-1 in §24. Lower retirements also mean fewer teardowns, so less used material and fewer green-time engines, and higher prices for both; that is row T-9. The shop-visit research describes the second effect from the shop's side: "the supply of used serviceable material remains thin, leaving shops more reliant on newly manufactured core parts". [AVM (Aviation Maintenance magazine), 2025, via D2 §2.4] The used-material research describes it from the trader's side, with AerSale reporting a "hypercompetitive feedstock environment" and its whole-asset sales falling from 20 engines and one aircraft in 2024 to 13 engines in 2025. [AerSale, 5 Mar 2026, as relayed in search results; D3 §2.6] Part V §33 carries the supply side in full.

20. New-generation engines: GTF and LEAP durability, and what they did to the old fleet

The A320neo and the 737 MAX (the re-engined 737) were designed to replace the A320ceo and the 737NG. Two things slowed the replacement. The new engines spend less time on wing than planned, time on wing being the interval from an engine's installation to its first removal (Part I §5). And the airframers delivered fewer aircraft than planned (§21). Both keep older aircraft flying longer and harder. This section takes the engines. Part I §10.2 boxed the LEAP (CFM's new-generation engine, the LEAP-1A on the A320neo family and the LEAP-1B as the sole

engine of the 737 MAX) and its durability kit; this section carries what each new engine's problem did to the old fleet, which Part I left to it.

Two new engines are involved, and they are not symmetric. The LEAP comes from the same manufacturer as the CFM56 and sits on both airframe families. The other comes from Pratt & Whitney and sits on one.

GEARED TURBOFAN (GTF)

The geared turbofan is Pratt & Whitney's new-generation engine. A reduction gearbox between the fan and the low-pressure turbine lets each turn at its own best speed, which the CFM56's direct-drive design does not allow. The PW1100G is one of two engine options on the Airbus A320neo family, the other being the LEAP-1A; the PW1500G and PW1900G are the sole engines of the Airbus A220 and the Embraer E2. Pratt & Whitney's parent is RTX. [D9 §2.5]

20.1 THE PRATT & WHITNEY GTF POWDER-METAL ISSUE

GTF POWDER-METAL ISSUE

Certain turbine discs in the GTF are made by a powder-metallurgy route, in which metal powder is consolidated into a disc. Powder produced over a period of years was found to be contaminated. The affected discs must be inspected or replaced, which can be done only with the engine off the wing and opened. RTX disclosed the defect in July to September 2023, estimated that 600 to 700 engines would be pulled for inspection, took a charge of about \$3 billion on 11 September 2023, and later estimates put the total programme cost at about \$7 billion. [D9 §2.5; BusinessDay, 11 Sep 2023; Mobility Outlook, date not captured]

The planning figure and the outturn diverged. RTX's planning figure at the time of disclosure was that around 350 aircraft on average would be grounded across 2024 to 2026. [The Flying Engineer, citing RTX, 2026] Cirium data cited in trade press put 835 PW1100G-powered aircraft grounded or in storage globally as of late 2025. [Cirium via The Flying Engineer, 2026] The same source gives "720 out of 1,912 A320neo family aircraft, nearly four in ten, currently out of service due to engine issues"; the denominator is not defined in the material reached, and the research judged it most plausibly to be the Pratt & Whitney-powered A320neo sub-fleet rather than all A320neos. [D9 T7, U5] An aircraft unable to fly for lack of a serviceable part or engine is said to be on the ground, abbreviated AOG (aircraft on ground), a term Part II §14 carries because short-term engine leases exist to cover such events.

RTX's own framing at its full-year 2025 results on 27 January 2026 was that 2026 is "the third year of elevated AOG levels"; that AOG levels were down more than 20% from their 2025 peak; that PW1100 AOG was down 25% year to date, enabled by a 40% increase in maintenance output and a 23% reduction in turnaround time; that fourth-quarter 2025 GTF maintenance

output was up 39% year on year; and that customer-compensation cash outflows were about \$1 billion in 2025, about \$700 million expected in 2026, and about \$2.8 billion cumulative by the end of 2026. [RTX FY2025 results via Leeham, 27 Jan 2026] Trade commentary expects the issue to be fully worked through only in 2027 to 2028. [The Flying Engineer, 2026] Named effects include Turkish Airlines grounding more than 40 aircraft and Wizz Air signing a compensation agreement with Pratt & Whitney. [Aviacionline; Shares Magazine; dates not captured]

EXHIBIT 3.7 — GTF GROUNDED-AIRCRAFT COUNTS AND RECOVERY INDICATORS, BY SOURCE AND DATE

Item	Figure	Source and level	Date
Engines to be pulled for inspection, initial estimate	600–700	RTX via BusinessDay (OEM via press)	11 Sep 2023
Charge taken / total programme cost	About \$3bn / about \$7bn	RTX via BusinessDay; Mobility Outlook	11 Sep 2023; later
RTX planning figure, aircraft grounded on average 2024–26	About 350	RTX, as cited by The Flying Engineer	2023 figure
PW1100G-powered aircraft grounded or in storage, late 2025	835	Cirium via The Flying Engineer (data provider via blog)	Late 2025
"Out of service due to engine issues"	720 of 1,912, about 38%; denominator undefined	The Flying Engineer	2026
AOG level against the 2025 peak	Down more than 20%	RTX FY2025 results via Leeham News	27 Jan 2026
PW1100 AOG, year to date	Down 25%	Same	27 Jan 2026
GTF maintenance output / turnaround time	Output up 40%; turnaround down 23%; Q4 output up 39% year on year	Same	27 Jan 2026
Customer-compensation cash	About \$1bn (2025); about \$700m (2026); about \$2.8bn cumulative by end-2026	Same	27 Jan 2026
Turkish Airlines aircraft grounded	More than 40	Aviacionline (trade press)	Date not captured

Item	Figure	Source and level	Date
GTF Advantage certification	EASA (the European Union Aviation Safety Agency) certified 2026; entry into service later in 2026	Aviation Today	2026
Expected full resolution	2027–28	The Flying Engineer (trade commentary)	2026

Source: BusinessDay, "RTX takes \$3bn charge over Pratt & Whitney jet engine recall", 11 Sep 2023; Mobility Outlook, "Troubled GTF engines to cost about \$7bn"; The Flying Engineer, "A320neo grounded 2026: Pratt & Whitney crisis" (citing Cirium); Leeham News, "RTX 2025 Earnings", 27 Jan 2026; Aviaconline; Aviation Today, GTF Advantage item. D9 §2.5 and §4.3; D9 T6 and T7.

The counts in that exhibit are in tension and the research did not resolve them. RTX's 2023 planning figure of about 350 aircraft grounded on average is less than half of Cirium's late-2025 count of 835 grounded or in storage. [D9 T6] The two are not the same measure: an average over three years is not a point count, and "in storage" may include aircraft parked for reasons other than the inspection. RTX's January 2026 statement gives a change, down more than 20% from the peak, without the level, so the peak itself and a year-by-year series from RTX's own disclosures were not reached. [D9 U5] The "720 of 1,912" figure cannot be reconciled with 835 without the article that gave it. [D9 T7] The primer carries all three.

GTF ADVANTAGE

The GTF Advantage is Pratt & Whitney's upgraded PW1100G, certified by EASA in 2026 for entry into service later in 2026. It is the fix for the fleet going forward; the discs already in service are dealt with by the inspection programme. [Aviation Today, 2026; D9 §2.5]

The mechanism that connects a grounded A320neo to the CFM56 aftermarket is replacement flying. Every PW1100G-powered A320neo on the ground needs its schedule flown by something else. The practical replacement is an A320ceo. The two share a pilot type rating, the licence endorsement that qualifies a pilot on one aircraft type, the same cabin-crew qualification and the same maintenance-organisation approvals, so an airline can substitute one for the other without retraining or re-approval. [D9 §2.5] That is why the A320ceo lease-extension and used-rent figures in §22 are what they are. The engines affected are the A320ceo's, the V2500 and the CFM56-5B, and not the 737NG's -7B.

WORKED EXAMPLE 3.3 — WHAT THE GTF GROUNDINGS MEAN IN A320CEO FLYING AND IN CEO ENGINE SHOP VISITS

Inputs. Grounded aircraft: 835 PW1100G-powered A320neos grounded or stored in late 2025 (sourced: Cirium via trade press); 350 aircraft on average in 2024–26 (sourced: RTX's 2023 planning

figure). Engines per aircraft: two. Utilisation of a mature narrowbody: 1,000–1,500 cycles a year (sourced: Aircraft Value News, via Part I §1). Mature CFM56 restoration interval: 10,000–15,000 cycles (sourced: Leeham, 8 Mar 2024, Part I §6.2); the V2500's interval was not retrieved, and this example applies the CFM56 interval to the whole ceo fleet as an assumption. Assumption: each grounded neo's flying is replaced one-for-one by a ceo that would otherwise have been parked or retired, at the same utilisation. That is the upper-bound case; in practice some flying was cancelled, some was absorbed by existing ceos flying harder (row T-12 in §24), and some by other types.

Step 1, engines whose flying must be replaced.

on 835 aircraft: $835 \times 2 = 1,670$ engine positions
on 350 aircraft: $350 \times 2 = 700$ engine positions

Step 2, cycles a year flown by the replacing ceo engines that would not otherwise have been flown.

on 1,670 positions: $1,670 \times 1,000$ to $1,670 \times 1,500 = 1.67$ to 2.5 million cycles a year
on 700 positions: $700 \times 1,000$ to $700 \times 1,500 = 0.7$ to 1.05 million cycles a year

Step 3, shop visits those cycles bring forward, at one restoration per 10,000–15,000 cycles.

on 1.67–2.5 million cycles: $1,670,000 \div 15,000 = 111$ to $2,500,000 \div 10,000 = 250$ visits a year
on 0.7–1.05 million cycles: $700,000 \div 15,000 = 47$ to $1,050,000 \div 10,000 = 105$ visits a year

Step 4, scale. The ceo-engine visits brought forward are 47 to 250 a year against a CFM56 forecast of 2,300–2,400 a year and a V2500 outturn of slightly more than 800 in 2025 (§23). They fall on the -5B and the V2500 in proportion to the ceo fleet mix, which is itself unknown (§18.2). Nothing in this example touches the -7B.

So the groundings add, at most, a few hundred ceo engine shop visits a year, and they do so by bringing visits forward in calendar time rather than by creating engines. The same engines would have reached those visits later, or been retired before them. Limits of the example: one-for-one replacement is the ceiling; the CFM56 interval is applied to the V2500 without a source; the 835 count includes stored aircraft that may be parked for other reasons (D9 T6); and the groundings were already falling by January 2026 (Exhibit 3.7), so the figures describe 2024–25 better than 2026.

[Cirium via The Flying Engineer, 2026; RTX via The Flying Engineer; Aircraft Value News, undated; Leeham News, 8 Mar 2024; D9 §2.5; reconciliation R-008]

20.2 LEAP DURABILITY AND THE JULY 2026 KIT

The LEAP-1A and -1B have had shorter time on wing than the CFM56 they replaced, particularly in hot and harsh environments such as the Middle East and India, driven by high-

pressure turbine (HPT) blade and fuel-nozzle issues. [ePlaneAI, two articles, dates not captured; D9 §2.5] Part I §10.2 recorded operators reporting removals at 2,000 to 6,000 cycles with "a planning base of 4,000", against a mature CFM56's restoration interval of 10,000 to 15,000 cycles. [Visual Approach, undated; Leeham News, 8 Mar 2024] The HPT is the hottest rotating part of the core and the usual limiter of time on wing (Part I §5).

The fix is the Durability Improvement Package (DIP), which Part I §10.2 boxed as the LEAP durability kit: a new HPT stage-1 blade, a new HPT stage-1 nozzle and a forward inner nozzle support. On 18 July 2026 CFM announced certification of the LEAP-1B HPT durability kit by the US Federal Aviation Administration (FAA) and by EASA, stating that it "will double time on wing in hot and harsh environments", that industrialisation was beginning, and that full production cutover is expected in early 2027. [CFM International / Safran / GE Aerospace release, 18 Jul 2026] The same release notes initial engine-level certification of a second change for the LEAP-1B.

REVERSE BLEED SYSTEM (RBS)

The reverse bleed system is a LEAP cooling arrangement that reduces the need for on-wing fuel-nozzle replacements, which CFM describes as "bringing on-wing maintenance needs in line with CFM56 standards". It received initial engine-level certification for the LEAP-1B alongside the HPT durability kit on 18 July 2026, and was already being deployed on the LEAP-1A. [CFM International release, 18 Jul 2026; D9 §2.5]

The LEAP-1A was ahead of the -1B. The new HPT blades and the reverse bleed fix were already being deployed on A320neo-family aircraft, with approximately one-third of the LEAP-1A fleet updated at the time of the report. [ePlaneAI, date not captured] By May 2026 GE Aerospace was stating that LEAPs "being shipped now will match the durability of the venerable CFM56". [Leeham News, 27 May 2026] Both the problem and the claimed fix are recorded; whether delivered LEAPs reach CFM56-level time on wing in service is not something the research could observe.

Scale and contract structure differ from the CFM56. About 5,500 LEAPs were in service at the date of one trade-press piece, with about 50% covered by long-term service agreements against about 15% of the CFM56 fleet. [Aviation Week, undated]

LONG-TERM SERVICE AGREEMENT (LTSA)

A long-term service agreement is a multi-year contract under which the airline pays a fixed amount per engine flight hour and the manufacturer or shop commits to perform or pay for the shop visits, taking the volume and cost risk and controlling the workscope, the parts sourcing and the shop. The research's dossiers call it an LTSA, an LTA, a rate-per-flight-hour agreement or a flight-hour agreement; one concept, several names. It matters here because about half the LEAP fleet is under one and about 15% of the CFM56 fleet is, so a LEAP shop visit mostly flows to the manufacturer's network while a CFM56 visit is mostly bought transactionally. Part IV §28 carries the full treatment.

[D2 §2.3; D9 §2.5; registry C-12]

CFM has outlined investments to support growth in LEAP shop visits, and Safran opened a new maintenance shop in Querétaro, Mexico, on 30 June 2026. [CAPA (Centre for Aviation); Safran release, 30 Jun 2026]

EXHIBIT 3.8 — LEAP DURABILITY: THE SHORTFALL, THE FIXES AND THEIR DATES

Item	Figure or date	Source and level	Date
Early LEAP-1A/-1B removals	2,000–6,000 cycles; planning base 4,000	Visual Approach (research house)	Undated
Mature CFM56 restoration interval, for contrast	10,000–15,000 cycles	Leeham News (trade press)	8 Mar 2024
Where the problem concentrates	Hot and harsh environments; HPT blade and fuel nozzle	ePlaneAI (trade press)	Dates not captured
LEAP-1B HPT durability kit certified (FAA and EASA)	18 Jul 2026; "will double time on wing in hot and harsh environments"	CFM / Safran / GE release	18 Jul 2026
LEAP-1B kit full production cutover	Early 2027	Same	18 Jul 2026
LEAP-1B reverse bleed system	Initial engine-level certification	Same	18 Jul 2026
LEAP-1A fleet updated with new HPT blades and RBS	About one-third	ePlaneAI	Date not captured
Manufacturer statement on delivered engines	LEAPs "being shipped now will match the durability of the venerable CFM56"	GE Aerospace via Leeham News	27 May 2026
LEAP in service / under long-term agreements	About 5,500 / about 50% (CFM56: about 15%)	Aviation Week	Undated
LEAP deliveries	1,802 (2025) against 1,407 (2024); Safran backlog >12,900	GE 10-K FY2025; Safran, 13 Feb 2026	Feb 2026

Item	Figure or date	Source and level	Date
New LEAP-capable shop	Safran Querétaro, Mexico, opened	Safran release	30 Jun 2026

Source: Visual Approach, "Wave of early LEAP-1A/1B shop visits frustrating operators", undated; Leeham News, Bjorn's Corner Part 49, 8 Mar 2024, and 27 May 2026; ePlaneAI, two articles; CFM International / Safran / GE Aerospace, "CFM secures certification for LEAP-1B durability upgrades", 18 Jul 2026; Aviation Week, "Leap Aftermarket Poised For Growth", undated; GE 10-K FY2025; Safran, 13 Feb 2026; Safran, Querétaro release, 30 Jun 2026. D9 §2.5 and §4.3; D1 §2.10.

The mechanism for the CFM56 aftermarket is indirect, and that is the point of recording it. A LEAP that comes off wing early needs a shop visit in a LEAP-capable shop, mostly within the manufacturer's network under a long-term agreement, and a spare LEAP while it is away. It does not generate a CFM56 shop visit, a CFM56 module or any CFM56 used material. Its effect on the old fleet runs through the gate described in §19.1: an airline whose A320neo or 737 MAX fleet is less available than planned slows its A320ceo or 737NG retirements. Willis Lease's FY2025 10-K lists "shorter on-wing engine time of most new technology engines" alongside airframer and engine-maker delays as compounding factors in the tight market. [Willis Lease Finance 10-K FY2025]

The timing of the fix bears on the shape of the CFM56 forecast, and here the research holds two readings. Safran's earlier framing, from about 2023 or 2024, had CFM56 shop visits peaking at about 2,500 a year in 2025 to 2026. GE's 2026 framing has a plateau of 2,300 to 2,400 a year through 2028 with the apex around 2027 to 2028 and a gradual fade after. [Safran via Aviation Week, c. 2023/24; GE via Aviation Week, 2026; reconciliation R-006, R-007] The LEAP-1B kit's full cutover in early 2027 is the date from which, on CFM's own statement, new 737 MAX engines should stop coming off wing early, and so the date from which this indirect support for the 737NG fleet begins to fade. That date sits inside GE's plateau and after Safran's earlier peak. The primer records the two curves and the cutover date beside each other and does not say which curve the cutover favours; §23.1 presents both curves in full.

20.3 WHICH OLD ENGINE EACH NEW-ENGINE PROBLEM TOUCHES

The two problems do not land on the same engines, and the table below is the map the translation table in §24 relies on.

EXHIBIT 3.9 — WHICH OLD ENGINE EACH NEW-ENGINE PROBLEM TOUCHES, AND THROUGH WHAT MECHANISM

New-engine problem	New airframe affected	Old airframe that substitutes	Old engines whose demand changes	Mechanism	Status at research date
GTF powder metal (PW1100G)	A320neo family, Pratt & Whitney-powered sub-fleet	A320ceo family	V2500 and CFM56-5B	Replacement flying on a shared type rating; ceo extensions, utilisation and shop visits brought forward	Groundings falling from the 2025 peak; resolution expected 2027–28
LEAP-1A durability	A320neo family, CFM-powered sub-fleet	A320ceo family	V2500 and CFM56-5B	Slower ceo retirement by operators with under-available neos	About one-third of the -1A fleet upgraded
LEAP-1B durability	737 MAX (sole engine)	737NG	CFM56-7B	Slower 737NG retirement by operators with under-available MAX fleets	Kit certified 18 Jul 2026; production cutover early 2027

Source: D9 §2.5 and §5; CFM International release, 18 Jul 2026; The Flying Engineer, 2026 (citing Cirium); ePlaneAI; Willis Lease Finance 10-K FY2025. Compiled for this Part.

So the GTF problem is an A320ceo story and, in engine terms, mostly a V2500 story with a CFM56-5B share. The LEAP-1B problem is a 737NG story and so a CFM56-7B story. Any sentence that attributes 737NG demand to the GTF, or A320ceo demand to the 737 MAX production cap, has the mechanism wrong. The translation table keeps them apart.

21. Airframer output: Boeing and Airbus

The second half of the replacement shortfall is airframes. Even where the new engines performed, there were fewer new aircraft to put them on than the airlines had ordered and planned for. This section gives the production and delivery figures for the two airframers with dates, and the arithmetic that connects a monthly rate to a year of undelivered replacements.

PRODUCTION RATE CAP (737 MAX)

The production rate cap is the monthly ceiling on 737 output imposed by the FAA after the Alaska Airlines door-plug event of January 2024: 38 aircraft a month, raised to 42 in October 2025, with the review to move to 47 completed in May 2026. Boeing's 2018 peak rate was 52 a month. The cap sets the maximum number of 737NG replacements that can be built in a year, and so the minimum number of 737NGs that cannot yet be released. [D9 §2.6; TravelPulse, Oct 2025; CAPA, May 2026]

BACKLOG

Backlog is the count of firm orders not yet delivered. Airbus's stood at a record 8,754 aircraft at the end of December 2025, more than ten years of production at 2026 delivery rates; the 737 MAX backlog exceeds 4,800 aircraft. A large backlog means that each aircraft not delivered this year is still owed, so the replacement is deferred rather than cancelled. [D9 §2.6; All About Industries; CAPA, May 2026; AirInsight]

GLIDER

A glider is a completed airframe parked at the factory awaiting engines. Airbus had about 60 at the end of the first half of 2025 and 32 at the end of the third quarter, and its chief executive said the number would reach zero by year-end. Gliders are the visible form of an engine-limited production line: the airframe exists, the delivery does not. [Leeham News, 30 Oct 2025; Air Data News; D9 §2.6]

21.1 BOEING AND THE 737 MAX

After the January 2024 door-plug event the FAA capped 737 production at 38 aircraft a month. [TravelPulse, Oct 2025] The FAA raised the cap to 42 a month in October 2025. [TravelPulse, Oct 2025] In May 2026 Boeing's chief executive, Kelly Ortberg, said the company had completed what he called the "capstone" review with the FAA, the final review in the agency's sequence, to move from 42 to 47 a month, and that the stabilisation at 42 had not taken long. [CAPA, May 2026] Boeing's stated plan was a further step in September 2026 and another in late 2026, potentially reaching 52 to 53 a month by year-end; Ortberg said he remained cautious about the step to 52 because of inventory concerns. [Travel Extra; AirInsight; CAPA, May 2026]

Deliveries followed the cap with a lag. 737 MAX deliveries were 387 in 2023, 260 in 2024 and 440 in 2025, and the cumulative total reached 2,399 by the date of the compilation the research used. [Wikipedia, 737 MAX orders and deliveries, 2026] Boeing's total deliveries of all types were 348 in 2024; by August 2025 it had delivered 385 aircraft, already more than in all of 2024. [Air Data News, Sep 2025] The compilation notes that Boeing exceeded 38 deliveries in some months by delivering from stored inventory, which is why 2025 deliveries exceed a full year at the capped build rate. [Wikipedia, 2026]

WORKED EXAMPLE 3.4 — WHAT A MONTHLY RATE CAP MEANS IN 737NG REPLACEMENTS A YEAR

Inputs (all sourced from §21.1): caps of 38, 42 and 47 a month and a 2018 peak of 52; MAX deliveries of 260 in 2024 and 440 in 2025. Assumption: every MAX delivered replaces one 737NG. That is the ceiling, since part of the backlog is growth rather than replacement (Cirium's Global Replacement Ratio of 46.1% in §19.2 says fewer than half of deliveries replace a retirement).

Step 1, a full year at each rate.

$$38 \times 12 = 456 \qquad 42 \times 12 = 504 \qquad 47 \times 12 = 564 \qquad 52 \times 12 = 624$$

Step 2, the delivery shortfall against the capped rate.

$$\begin{aligned} 2024: & 456 - 260 = 196 \text{ aircraft short of a full year at 38 a month} \\ 2025: & 440, \text{ roughly a full year at 38 plus deliveries from storage} \end{aligned}$$

Step 3, the gap between the 2018 rate and the cap, as a rough measure of the replacement that did not happen for production-cap reasons alone.

$$(52 - 38) \times 12 = 168 \text{ aircraft a year, or 336 CFM56-7B engine positions not released}$$

So the cap alone held back on the order of 170 737NG replacements a year in 2024 and 2025, before counting the 2019–20 grounding and the 2021–23 ramp. At the replacement ratio of 46.1%, the figure would be about 77 aircraft and 155 engines. Limits of the example: the one-for-one assumption is the ceiling; the replacement ratio is a global all-type figure, not a 737 figure; and the cap is lifting, so the figure shrinks each year. [D9 §2.6, arithmetic on sourced figures; Wikipedia, 2026; CAPA, May 2026; Cirium via Informa]

21.2 AIRBUS AND THE A320NEO FAMILY

Airbus originally targeted 820 deliveries in 2025, later guided to about 790, and delivered 793. [Argus Media; Air Data News; D9 T9] Both the original and the revised targets are recorded, since the figure was revised during the year. It targets around 870 deliveries in 2026; in August 2026 it delivered 57 aircraft, described as leaving "a steep climb" to that target. [Air Data News; Aerocorner, Sep 2026] The engine shortfall behind the gliders came mainly from Pratt & Whitney, whose powder-metal production and inspection load competed with new-engine output, and earlier in 2026 from CFM, which fell behind before catching up. [Air Data News] Airbus was still negotiating Pratt & Whitney engine volumes for 2026 production in late 2025. [Air Data News]

The rate ambition moved. Airbus now expects A320neo-family output to reach 70 to 75 a month by the end of 2027 and to stabilise at 75 thereafter, instead of reaching that rate sooner as earlier plans had it. [Air Data News; Mexico Business News] The backlog was a record 8,754 aircraft at the end

of December 2025. [All About Industries] The A320 family, ceo and neo combined, had 20,378 orders, 12,798 deliveries and 11,308 aircraft in service as of July 2026; the ceo/neo split of the in-service figure was not reached. [Wikipedia, July 2026; D9 U1]

EXHIBIT 3.10 — AIRFRAMER OUTPUT 2023–2027: CAPS, DELIVERIES, GLIDERS, RATES AND BACKLOGS, WITH DATES

Item	Figure	Source and level	Date
737 production cap imposed	38 a month	FAA, via TravelPulse (trade press)	Jan 2024
737 cap raised	42 a month	FAA, via TravelPulse	Oct 2025
Review completed to move to 47	47 a month	Boeing CEO via CAPA	May 2026
Boeing plan for late 2026	52–53 a month; cautious about 52	Travel Extra; AirInsight; CAPA	2026
2018 peak rate	52 a month	CAPA; AirInsight	2018 figure
737 MAX deliveries	387 (2023); 260 (2024); 440 (2025)	Wikipedia compilation	2026
737 MAX cumulative deliveries / backlog	2,399 / more than 4,800	Wikipedia; CAPA	2026
Boeing total deliveries, all types	348 (2024); 385 by Aug 2025	Air Data News	Sep 2025
Airbus 2025 target, original / revised / actual	820 / about 790 / 793	Argus Media; Air Data News	2025
Airbus 2026 target	About 870; 57 delivered in Aug 2026	Air Data News; Aerocorner	Sep 2026
Airbus gliders	About 60 (end 1H 2025); 32 (end Q3 2025); zero targeted (end 2025)	Leeham News; Air Data News	30 Oct 2025
A320neo-family rate ambition	70–75 a month by end-2027; 75 thereafter	Air Data News; Mexico Business News	2026
Airbus backlog	8,754 aircraft	All About Industries	End-Dec 2025
A320 family (ceo and neo) orders / deliveries / in service	20,378 / 12,798 / 11,308	Wikipedia compilation	July 2026

Source: TravelPulse, "FAA Raises Boeing 737 MAX Production Cap", Oct 2025; CAPA, "Boeing testing production rate of 47 aircraft per month", May 2026; Travel Extra; AirInsight; Wikipedia, "List of Boeing 737 MAX orders and deliveries" and "List of Airbus A320 family orders and deliveries"; Air Data News, three articles; Argus Media; Aerocorner, Sep 2026; LeeHam News, "Airbus 3Q2025", 30 Oct 2025; Mexico Business News; All About Industries. D9 §2.6 and §4.3; D9 T9.

21.3 WHAT AIRFRAMER OUTPUT DOES TO THE OLD FLEET

The mechanism is the gate of §19.1 again, seen from the factory. Each A320neo or 737 MAX not delivered in a year is an A320ceo or 737NG that its airline cannot release that year. The airline that waits for a late new aircraft still pays for the old one's maintenance in the meantime. So a delivery shortfall converts the airline's planned capital spending on a new aircraft into aftermarket operating spending on an old one, for as long as the delay lasts. Because the backlogs are owed rather than cancelled, the conversion is a deferral: when the rate rises, from 42 toward 52 at Boeing and toward 75 at Airbus, the deferred replacements arrive and the deferred retirements with them. That is why the translation table's rows for the two airframers (T-4 and T-5 in §24) carry a reversal in their third column. Air Lease's FY2025 10-K puts the lessor's view of the same mechanism in one sentence: "As global air traffic expands and aircraft production volumes remain constrained, lessors are experiencing strong demand for aircraft through new lease requests and lease extension requests." [Air Lease Corp 10-K FY2025]

22. Lessors and airlines: extensions, availability, lease rates

The owners of the old fleet responded to the closed gate in ways that show up directly in aftermarket demand. This section takes the airline's maintenance decision at fleet level, then the decision to extend or return a leased aircraft, then what the lessors did between 2023 and 2026 and what it did to lease rates. Part II owns the operating lease (§12), maintenance reserves and return conditions (§13), engine leasing (§14) and green time (§16); this section uses those terms and adds the context.

22.1 THE AIRLINE'S MAINTENANCE DECISION AT FLEET LEVEL

An airline's engineering and fleet-planning functions do not wait for engines to fail. They run a rolling forecast of when each engine will have to come off.

ENGINE REMOVAL FORECAST

The engine removal forecast is an airline's rolling schedule of each installed engine's projected date of removal for a shop visit. Each engine's date is set by whichever of its limits is reached first: remaining exhaust-gas-temperature margin (Part I §5), remaining life-limited-part cycles (Part I §4), or a hard inspection. From the forecast the airline books shop slots, in 2025 and 2026 some 12 to 18 months ahead, because the wait for a slot runs two to six months. Part IV §27 owns the slot and induction. [D9 §2.7; Air Cargo Week, 2026; Locatory, Mar 2026]

Which of the three clocks runs out first is itself contested in the research, and Part I §5.2 carries the tension. Older trade-press analyses found exhaust-gas-temperature margin erosion "generally the most influential factor" in short-haul removal intervals; later analyses of the latest -5B and -7B variants found margin lasting to the life-limited-part limit, so that removals are LLP-driven, with "especially first-run engines" flown almost to the limit. [Aircraft Commerce, 2003 and 2008; Aircraft Commerce Issue 50; FlightGlobal; reconciliation R-011] The demand research lists the three clocks without ranking them, and so does this Part.

For each removal the airline chooses a workscope (Part I §6), meaning how much of the engine to open and restore. The choice depends on how long the airline expects to keep the aircraft. If the aircraft will be retired or returned to its lessor within a few years, the airline avoids a full performance restoration and buys green time instead: it installs an engine or module with enough margin and LLP life to reach the retirement date, then returns or sells it at or near the end of its life. The green-time engine comes from the used-material market or from an engine lessor. Trade press describes airlines "replacing LLP-expired modules with green-time units to maintain engines on wing while awaiting new aircraft or engine deliveries", through quick-turn or hospital shop visits that do only the essential work (Part I §6; Part VII §46). [Air Cargo Week, 2026; Aviation Business News] If the aircraft will stay for a long time, the airline does the full restoration and replaces the LLPs so the engine runs another full interval.

Green time is a finite pool, and the research records its limit. Bain's 2024 brief on the engine-maintenance capacity crunch notes that at 80% fleet utilisation green-time capacity becomes tight, and that "there is no huge buffer if airlines rely on green time and are getting to 80%". [Bain, 2024] The pool is fed by retirements. When retirements are low the pool shrinks exactly when demand for it is highest, which is the mechanism behind row T-9 of the translation table.

Spare engines are the other quantity the airline manages (Part II §14). An airline carries spares to cover removals; when shop turnaround lengthens, the same removal rate ties up more spares, and the airline buys or leases more. Two filings give the scale at airline level. Spirit Aviation Holdings had 16 spare engines financed under operating leases and owned 18 at 31 December 2025. [Spirit Aviation Holdings 10-K FY2025] American Airlines had agreements for 47 spare engines to be delivered in 2026 and beyond. [American Airlines 10-K FY2025]

22.2 THE EXTEND-OR-RETURN DECISION

When a lease expires the airline weighs three things. The first is the extension rent the lessor asks. The second is the cost of meeting return conditions (Part II §13), the contractual minimum remaining life on engines, LLPs and airframe checks at redelivery, which can force a shop visit just before return. The third is the availability and price of a replacement. [D9 §2.7] Between 2023 and 2026 the third has usually been decisive. With new aircraft late and used aircraft scarce,

extending has been the only way to keep capacity. So airlines extend and perform the maintenance the extended term needs, rather than returning aircraft at the end of their useful life and leaving the shop visit to the next owner. The lessor, facing the same scarcity from the other side, prefers extension to a transition it would have to pay for (§19.1).

22.3 HOW LESSOR BEHAVIOUR CHANGED, 2023–2026

LEASE EXTENSION AND THE EXTENSION SHARE

A lease extension renews an expiring operating lease with the incumbent airline instead of taking the aircraft back for transition, sale or part-out. The extension share is the proportion of scheduled lease expiries that are extended rather than returned. Cirium Ascend's head of consultancy, Rob Morris, said 75 to 90% of scheduled expirations were being extended; IBA found extension terms on legacy single-aisle aircraft stretching to four to six years, with extension rents in some cases more than 40% above the age-specific rates agreed just before the spike. Each extension keeps an aircraft and its two engines out of the used market and inside the maintenance cycle for the term. [Cirium Ascend via Bloomberg/AJOT, 2023–24 context; IBA via Key.Aero and IBA Weekly, Nov 2023; D9 §2.8]

The availability figures measure how far the market tightened. Cirium Ascend estimated that only about a dozen mid-life 737NGs and fewer than 25 A320ceos were available to rent anywhere in the world at the time of its count, and that lease rates on ten-year-old 737-800s were 44% above January 2022 levels. [Cirium Ascend via Bloomberg/AJOT; date not visible, context 2023–24] IBA's research found lease lengths being extended, terms growing to four to six years, extension deals running out to 2026 and beyond, and in extreme cases aircraft due back at year 12 not expected to return until 2032, at 18 years of age. [IBA via Key.Aero; IBA Weekly, 24 Nov 2023]

The lessors' own filings describe the same market. Air Lease's FY2025 10-K: "The strong demand for aircraft, combined with elevated interest rates and inflation, helped increase lease rates on new lease agreements and lease extensions during 2025. Lease rates are expected to remain strong as the supply and demand environment for commercial aircraft remains tight." [Air Lease Corp 10-K FY2025] Willis Lease's FY2025 10-K: "Both Airbus and Boeing have delivery delays which have been compounded by engine manufacturer delays, shorter on-wing engine time of most new technology engines, and ongoing supply chain challenges." [Willis Lease Finance 10-K FY2025]

EXHIBIT 3.11 — LEASE-MARKET INDICATORS 2022–2026, BY SOURCE, LEVEL AND DATE

Indicator	Figure	Source and level	Date
Share of scheduled lease expiries extended	75–90%	Cirium Ascend (Rob Morris) via Bloomberg/AJOT (data provider's consultancy via press)	Undated; context 2023–24
Mid-life aircraft available to rent worldwide	About 12 737NGs; fewer than 25 A320neos	Same	Same
Lease rate, ten-year-old 737-800	+44% against January 2022	Same	Same
Extension terms on legacy single-aisles	4–6 years; extension deals to 2026 and beyond; some aircraft due back at year 12 not returning until 2032, at age 18	IBA (appraiser) via Key.Aero; IBA Weekly	Nov 2023
Extension rents against prior age-specific rates	More than 40% above, in some cases	IBA	Nov 2023
Twelve-year-old 737-800 rent	About \$228,000 a month, down 11% year on year	IBA	Jul 2025
Twelve-year-old A320-200 rent	About \$220,000 a month, down 13% year on year	IBA	Jul 2025
Twenty-one-year-old 737-800 rent	Less than \$160,000 a month	IBA	Jul 2025
Narrowbody conversion and lease-rate outlook	Warning of "oversupply of narrowbody conversions and lease rate fall"	IBA via India Seatrade News	6 Sep 2026
New-aircraft lessor on 2025	Rates up on new leases and extensions; "expected to remain strong"	Air Lease Corp 10-K FY2025 (filing)	Feb 2026
Engine lessor on causes	Airframer delays compounded by engine-maker delays, shorter on-wing time, supply chain	Willis Lease Finance 10-K FY2025 (filing)	2026
CFM56-7B spare-engine rent	About \$100,000 a month in 2024, from about \$75,000 in 2019	IBA	Apr 2024

Indicator	Figure	Source and level	Date
CFM56-7B rent, alternative reading	\$42,000–48,000 a month, "about 34% above 2019", up 15–20% "in recent months"; may describe green-time engines	Safe Fly Aviation (consultancy blog)	2026
CFM56-7B half-life market value and its direction	\$5.7M (-7B24), \$6.4M (-7B27); values "levelled off after approximately 18 months of increases"	IBA	Sept 2025; H1 2026
Lauda Europe (Ryanair group) A320 leases	Extended through the early 2030s	Travel Extra (trade press)	2026
Spirit spare engines	16 under operating leases plus 18 owned	Spirit Aviation Holdings 10-K FY2025	31 Dec 2025
American Airlines spare engines on order	47 for 2026 and beyond	American Airlines 10-K FY2025	2026

Source: Bloomberg via AJOT, "Used jets rents surge 44% as airlines pay up to buoy fleets"; Key.Aero, "Lease trends" (IBA research); IBA Weekly, 24 Nov 2023; India Seatrade News, "IBA warns of oversupply of narrowbody conversions and lease rate fall", 6 Sep 2026 (carrying IBA's July 2025 commentary); Air Lease Corp, Form 10-K FY2025; Willis Lease Finance, Form 10-K FY2025; AviTrader, "IBA says it's a lessor's market", 25 Apr 2024; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); IBA, Engine Values Release 2025B, Sept 2025, and H1 2026 update via Air Cargo Update; Travel Extra, 2026; Spirit Aviation Holdings and American Airlines, Forms 10-K FY2025, D9 §2.8 and §4.4; D6 §2.1 and §4; reconciliation R-028, R-029, R-031, R-034.

Three tensions sit in that exhibit, and the primer carries each with both sides. The first is the direction of mid-life narrowbody lease rates. Air Lease, a lessor of new aircraft, reported in its FY2025 filing that rates rose on new leases and extensions in 2025 and were expected to remain strong, and Cirium's 44% rise on ten-year-old 737-800s points the same way. [Air Lease Corp 10-K FY2025; Cirium Ascend via Bloomberg] IBA, an appraiser, put a twelve-year-old 737-800 at about \$228,000 a month in July 2025, down 11% year on year, and a comparable A320-200 at about \$220,000, down 13%, and in September 2026 warned of a lease-rate fall. [IBA, Jul 2025 and 6 Sep 2026, via India Seatrade News; reconciliation R-031] The sources describe different asset ages at different dates, a new-aircraft lessor's whole book against an appraiser's mid-life commentary, and the primer does not reconcile them.

The second is freighter conversion. FTAI and the conversion house AEI announced a collaboration on 7 July 2026 to deliver "a more cost-effective Boeing 737-800 freighter solution", and the type-specialist data show more than 260 737-800 freighters in service with a pipeline above 300. [FTAI and AEI release, 7 Jul 2026; Acumen Aviation; Cargo Facts] IBA's September 2026

note warns of an "oversupply of narrowbody conversions". [IBA via India Seatrade News, 6 Sep 2026; reconciliation R-034] Part VIII §67 carries the collaboration; this Part records that the conversion demand the translation table relies on (row T-11) is contested by an appraiser at the research date.

The third is the engine lease rate, which Part II §14 owns. IBA put a CFM56-7B spare engine at about \$100,000 a month in 2024, up from about \$75,000 in 2019. The consultancy blog gives \$42,000 to \$48,000 a month in 2026, "about 34% above 2019", which the leasing research notes sits beside green-time values and may describe green-time engines, although the source does not say so. [IBA, Apr 2024; Safe Fly Aviation, 2026; reconciliation R-028] The percentage change from 2019 agrees across the two, about a third, while the levels do not. [reconciliation B-5]

WORKED EXAMPLE 3.5 — A 100-AIRCRAFT LEASE COHORT, AND WHAT AN EXTENSION DOES TO THE MAINTENANCE CALENDAR

Inputs. Cohort: 100 A320ceo leases expiring in one year (assumed, for scale). Extension share: 75–90% (sourced: Cirium Ascend). Extension term: 4–6 years (sourced: IBA, Nov 2023). Utilisation: 1,000–1,500 cycles a year per engine (sourced: Aircraft Value News). Mature restoration interval: 10,000–15,000 cycles (sourced: Leeham, Mar 2024). Stock available to rent: fewer than 25 A320ceos worldwide (sourced: Cirium Ascend).

Step 1, the aircraft that come back.

returns: $100 - 75 = 25$ aircraft at a 75% share; $100 - 90 = 10$ aircraft at 90%
engines returned with them: 20 to 50

Those 10 to 25 aircraft are the entire flow into the used market from that cohort for transition, sale or part-out. Step 2, the stock against the flow. Cirium's count of fewer than 25 A320ceos available to rent worldwide is of the same order as one year's returns from a single 100-aircraft cohort. That is the arithmetic behind a 40%-plus move in used rents: the stock was one cohort's returns deep (D9 §2.8).

Step 3, cycles added inside the extension.

$4 \text{ years} \times 1,000 = 4,000$ cycles, to $6 \text{ years} \times 1,500 = 9,000$ cycles per engine

Step 4, whether the next shop visit falls inside the term. Take an engine at expiry with 3,000 cycles of life remaining to its next restoration (assumed). Any extension in the range forces that restoration inside the term, and the incumbent airline pays for it rather than leaving it to redelivery. Take instead an engine with 10,000 cycles remaining (assumed). A four-year extension at 1,000 cycles a year, 4,000 cycles, avoids the visit; a six-year extension at 1,500, 9,000 cycles, nearly reaches it.

engines whose next event now sits inside an extended term: 75 to 90
aircraft $\times 2 = 150$ to 180 engines

So in a 100-aircraft cohort, extension moves the next maintenance event of 150 to 180 engines from a future owner's calendar into the incumbent's, and holds 20 to 50 engines back from the used and teardown market. Limits of the example: the cohort size is assumed; the remaining life at expiry is assumed and in practice is spread across the cohort; return conditions (Part II §13) may already require the visit at redelivery, in which case the extension changes who pays and when, not whether; and the V2500's interval is applied from the CFM56's. [Cirium Ascend via Bloomberg/AJOT; IBA via Key.Aero, Nov 2023; Aircraft Value News, undated; Leeham News, 8 Mar 2024; D9 §2.8]

22.4 FTAI'S POSITION IN THIS CHAIN

FTAI's FY2025 10-K describes the 2025 Partnership, the first vehicle of its Strategic Capital Initiative (Part VIII §60 to §62), as focused on acquiring 737NG and A320ceo aircraft and as the primary buyer of all future on-lease narrowbody aircraft. [FTAI Aviation 10-K FY2025] The lessor behaviour described above cuts two ways for a buyer of mid-life narrowbodies, and the research records both without weighing them. High extension rates and rents support the value of aircraft already owned. The same scarcity makes new acquisitions harder to source, because the 10 to 25 aircraft a year that come back from a 100-aircraft cohort are all there is to buy. [D9 §2.8] Part VIII carries the vehicle; Part II §15 carries why the engines carry the value of a mid-life narrowbody.

23. Forecasts: shop-visit curves and the engine MRO market

The quantities in §18 to §22 feed forecasts. This section sets out every forecast curve the research reached, the measure each uses, and its date, and then reconciles the headline shop-visit count against the installed base. Part IV §26 carries the market's reading of the same curve as "shop-visit demand"; the two sections cross-reference and do not duplicate.

SHOP-VISIT FORECAST AND FIRST SHOP VISIT

A shop-visit forecast counts the expected engine inductions per year for a type. A first shop visit is the first removal of an engine for shop maintenance since new; first runs are usually the longest, so a fleet's first-visit peak comes years after its delivery peak. The two measures can peak in different years: GE and Safran forecast total CFM56 visits of about 2,300 to 2,400 a year in 2026 to 2028, while an independent research house dates the peak in first visits for the CFM56 and V2500 to 2027. [D9 §2.9; GE via Aviation Week, 2026; Visual Approach Analytics]

MRO MARKET AND ENGINE MRO

MRO is maintenance, repair and overhaul, the generic term for aircraft and engine upkeep (Part IV §25). The MRO market is the dollar value of that spending in a year, and engine MRO is the engine share of it. Oliver Wyman put engine MRO at \$70 billion in 2026, rising to \$102.6 billion by 2036. Aviation Week counts the same market in events: about 9,000 engine shop visits of all types in 2025, about 2,300 of them CFM56. [Oliver Wyman via AviationPros and Air Cargo Week, 2026; Aviation Week 2026 Fleet & MRO Forecast via ePlaneAI; D9 §2.9]

UNCONDITIONAL MRO DEMAND

Aviation Week's 2026 forecast gives "unconditional MRO demand of \$853bn over ten years". The forecast's own definition was not retrieved. Read from first principles, unconditional demand is the maintenance the fleet would require if every scheduled event were performed when due, before any constraint of shop capacity, parts or labour is applied; it is the quantity Bain's capacity model is compared against. The primer uses the phrase only in that sense and flags that the publisher's definition is unverified. [Aviation Week 2026 Fleet & MRO Forecast via ePlaneAI; registry section 2]

23.1 THE CFM56 CURVE

The research reached the CFM56 curve from the two CFM partners at different dates, from a trade-press forecast, from an independent research house and from a consultancy blog. They do not describe one shape.

EXHIBIT 3.12 — CFM56 SHOP-VISIT FORECAST STATEMENTS BY SOURCE, MEASURE AND DATE

Period or measure	Figure	Source and level	Date
2019, the pre-downturn level	"Just above 2,000" CFM shop visits, "almost all of them CFM56-5B and -7B"	Safran via Aviation Week (OEM via trade press)	c. late 2023 / early 2024
2020–2023	Not found year by year	D2 U2	Not dated
2024	"Topping 2,000 again in 2024, with some help from the Leap family"	Safran via Aviation Week	c. 2023/24
2024 → 2025	"40 per cent increase in shop visits"; scope (all engines or CFM56) not captured	IBA via Aviation Week headline	2024
2025	CFM56 shop visits grew "mid-single-digit", with "a higher proportion of full work scope shop visits"	Safran FY2025 results release; Forecast International	13 Feb 2026
Peak, earlier framing	About 2,500 a year in 2025–26	Safran via Aviation Week	c. 2023/24

Period or measure	Figure	Source and level	Date
2026 and 2027	"Entered the year projecting about 2,300"; "expanded the range to 2,300–2,400"; "probably closer to 2,400"	GE Aerospace via Aviation Week	2026
Through 2028	2,300–2,400	Same	2026
Apex	"Around the 2027–2028 time period, followed by a gradual fade"	Same	2026
Removals, 1H 2027 against 2026	Up about 10%	GE Aerospace via Aviation Week	2026
First-shop-visit peak, CFM56 and V2500	2027	Visual Approach Analytics (research house headline)	2025/26
2030, all CFM engines	"More than 4,000 shop visits annually, roughly split between Leaps and previous-generation models", so about 2,000 CFM56	Safran via Aviation Week	c. 2023/24
LEAP curve	Shop visits more than double by 2030; peak above 5,000 a year by 2040; LEAP aftermarket revenue parity with CFM56 in 2028	Safran Capital Markets Day via Quartr	2024
737-700 passenger MRO demand	"To decline sharply after 2027"	Acumen Aviation	18 Dec 2025
Engine MRO demand, all types	"Near-term peak in 2026", constrained through the end of the decade	AviationPros (trade press)	2026
Consultancy reading	2,300–2,400 a year through 2028, labelled "CFM56-7B shop visits" by one dossier and all CFM56 by another	Safe Fly Aviation (consultancy blog)	2026

Source: Aviation Week, "Safran Sees Leap Aftermarket Emerging By Mid-Decade", c. 2023/24, and "Leap Aftermarket Poised For Growth", undated; Aviation Week, "IBA predicts 40 per cent increase in shop visits from 2024 to 2025", 2024; Safran, FY2025 results release, 13 Feb 2026; Forecast International, 13 Feb 2026; Aviation Week, "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; Visual Approach Analytics; Quartr, Safran Capital Markets Day 2024 summary; Acumen Aviation, 18 Dec 2025; AviationPros, Engine Stand Utilization Report; Safe Fly Aviation, 2026 (consultancy blog). D2 §4.1; D9 §2.9 and §4.5; reconciliation R-006, R-007; E.1 rule 15.

The level and the shape are both contested, and the reconciliation requires both readings wherever the plateau is discussed. [E.1 rule 15] On level, Safran's earlier framing put the peak at about 2,500 a year in 2025 to 2026; GE's 2026 framing entered the year at about 2,300 for each of 2026 and 2027, widened to 2,300 to 2,400, "probably closer to 2,400", and held that range

through 2028. [Safran via Aviation Week, c. 2023/24; GE via Aviation Week, 2026; reconciliation R-006] On shape, the earlier view is a peak and the later one a plateau with an apex around 2027 to 2028 and a gradual fade. [reconciliation R-007] Visual Approach's 2027 peak is in first visits only, a different measure, so it is not strictly contradictory, but it describes a different shape again. [Visual Approach Analytics; D9 T4] Two further figures are unreconciled with either curve. IBA's headline of a 40% increase in shop visits from 2024 to 2025 does not say whether it counts all engines or the CFM56, and Safran's own full-year 2025 statement has CFM56 visits up only "mid-single-digit", with more full worksopes. [IBA via Aviation Week, 2024; Safran, 13 Feb 2026; reconciliation R-006] The year-by-year counts for 2020 to 2023 that would let a reader see the recovery were not found. [D2 U2]

The long end of the curve depends on the retirement rate, which is why §19 came first. Safran's 2030 figure of more than 4,000 visits across all CFM engines, roughly half LEAP, implies about 2,000 CFM56 visits in 2030, and Safran's 2024 Capital Markets Day put LEAP aftermarket revenue at parity with the CFM56 by 2028. [Safran via Aviation Week, c. 2023/24; Safran Capital Markets Day 2024 via Quartr] Those are forecasts made when the retirement assumption was higher than the outturn, and the research does not say how they would move at 1.5 to 2%. [D9 T3]

23.2 THE V2500 CURVE

The V2500 figures are in direct contradiction and the primer carries both. Aviation Week's V2500 data tool projected a spike to about 1,400 V2500 shop visits in 2025 and more than 7,800 overhaul visits plus 3,142 LLP-related visits over the following ten years. [Aviation Week V2500 data tool, date not visible] RTX's chief financial officer, reported by the same publication in 2026, put 2025 at "slightly more than 800" and expected 2026 to be about the same, "probably within 20". [RTX CFO via Aviation Week, 2026; reconciliation R-009] A published forecast and a reported outturn, both via one publication, one quoting the manufacturer's finance chief. IAE expected "no retirement surge" for the V2500 over the next five years. [IAE via Aviation Week, 2024/25] MTU (MTU Aero Engines, the German engine maker whose maintenance arm is independent on this engine) performed 38% of all V2500 shop visits in 2024. [MTU Aero Engines, 2025] Part VII §51 carries FTAI's V2500 programme against this market.

23.3 THE MARKET IN DOLLARS AND IN VISITS

EXHIBIT 3.13 — ENGINE MRO MARKET SIZE, SHOP-VISIT COUNTS AND THE CAPACITY SHORTFALL, BY SOURCE AND DATE

Item	Figure	Source and level	Date
Engine shop visits, all types	About 8,800 (2024); about 9,000 (2025), of which about 2,300 CFM56	Aviation Week 2026 Fleet & MRO Forecast via ePlaneAI (trade press)	2026

Item	Figure	Source and level	Date
Ten-year MRO demand	\$853bn "unconditional"; aftermarket surpassing \$1.6 trillion cumulative at a 3.2% compound annual growth rate, as read	Same	2026
Engine MRO market	\$70bn (2026) rising to \$102.6bn (2036)	Oliver Wyman (consultancy) via AviationPros and Air Cargo Week	2026
Global MRO market, all segments	More than \$136bn (2025); about \$140bn (2026)	Oliver Wyman, "The new MRO supply paradigm"	Apr 2026
Narrowbody share of commercial MRO	58.3% (2025)	GM Insights (a market-research vendor; low grade)	2025
Engine MRO capacity shortfall	Demand to exceed available supply by more than 17% by the end of the decade if capacity grows on its current trajectory	Bain (consultancy)	2024
Engine shop visits to 2040	"Set to rise sevenfold"; base not stated	IATA study via AFM (a trade publication)	Undated
V2500 shop visits	Slightly more than 800 (2025); 2026 within about 20 of that	RTX CFO via Aviation Week	2026
V2500 shop visits, forecast	About 1,400 (2025); more than 7,800 overhaul plus 3,142 LLP visits over ten years	Aviation Week data tool	Date not visible

Source: ePlaneAI, "Forecast for US commercial fleet and MRO growth through 2026" (reporting Aviation Week 2026 Fleet & MRO Forecast figures); AviationPros, "6-Month Analysis: Top Engine Maintenance Trends for MROs to Watch in 2026" (Oliver Wyman figures); Air Cargo Week, "From PW1100G to CFM56", 2026; Oliver Wyman, "The new MRO supply paradigm", Apr 2026; GM Insights; Bain & Company, "Commercial aviation forecast" and "Get a step ahead of the engine maintenance capacity crunch", 2024; AFM; Aviation Week, "Pratt & Whitney Sees High-Single-Digit MRO Growth This Year", 2026, and V2500 data tool. D9 §2.9 and §4.5; D2 §4.7; reconciliation R-009; D9 T10.

Two cautions attach to that exhibit. The IATA figure of a sevenfold rise in shop visits by 2040 has no stated base; a sevenfold rise from about 9,000 all-engine visits would be about 63,000, which no manufacturer's curve supports, so the research judged the base to be new-generation engines only and treats the figure with caution until the study is read. [AFM; D9 T10] Bain's shortfall of more than 17% is conditional: it holds if engine MRO capacity keeps growing only on its current trajectory, and Bain's own brief describes the capacity responses under way. [Bain, 2024] The type-level curve for the CFM56 through 2030 inside Aviation Week's forecast, and

Oliver Wyman's narrowbody-engine subtotal by year, were not reached; only the headline totals surfaced. [D9 U7]

23.4 CAPACITY, TURNAROUND AND STANDS

Part IV §27 owns capacity. Four of its figures are needed here because the translation table's row T-8 depends on them, and because the research holds two readings of whether the constraint is easing. Trade press puts a full CFM56 overhaul at 90 to 120 days against about 60 before the pandemic, light work at about 45 days, legacy-engine turnaround times up 35% and new-generation times up more than 150% against five years earlier, and slot waits at two to six months. [Air Cargo Week, 2026; Aviation Business News / AVM, 2025] GE's own filing and conference material read the other way: turnaround for the LEAP, CFM56 and GE90 improved by more than 10% year on year in the fourth quarter of 2025, material input from priority suppliers rose more than 40%, and CFM56 turnaround was "sustaining about 90 days" in May 2026. [GE 10-K FY2025; GE Aerospace, Bernstein conference, 27 May 2026; reconciliation R-019, R-020] Oliver Wyman's April 2026 survey has two-thirds of MRO operators unable to find enough technicians, labour-rate inflation at 5.5 to 6.0%, and more than half expecting attrition to worsen. [Oliver Wyman, Apr 2026] A filing and a conference deck against trade press and a consultancy survey; the primer carries both, and Part IV §27 works them through.

ENGINE STAND

An engine stand is the wheeled fixture that holds an engine or a module during work, storage and transport. Because an engine off the wing must sit on one, stands are a physical limit on work in progress, and their utilisation is a measure of how full the system is: legacy-engine stand use rose by a third in 2025 over 2024, and CFM56-5A/-5B stand use rose 60%. Part IV §27 carries the full treatment. [AviationPros, Engine Stand Utilization Report; D2 §2.4; D9 §6]

Third-party test and repair waits of four to six months, and advice to buyers to plan procurement early for LLPs and critical rotables and to use exchange pools, used material and diversified suppliers, complete the picture the trade press gives. [Locatory, Mar 2026; AVM, 2025] Aero Norway, an independent CFM56 shop, said the long lead times for induction slots were driving interest in quick-turn services, "especially in the case of older-generation engines, such as the CFM56 and V2500". [Aero Norway in AVM, May 2024]

WORKED EXAMPLE 3.6 — RECONCILING 2,300 SHOP VISITS A YEAR WITH ABOUT 24,000 ENGINES

Inputs. Visits: 2,300 a year, the low end of GE's 2,300–2,400 (sourced: GE via Aviation Week, 2026). Bases (sourced, Exhibit 3.1): about 24,000 (trade press, Sept 2025); more than 22,800 (Safran, end-2025); more than 19,000 "flying" and about 14,200 "active" (consultancy blog, labelled). Interval: a mature performance restoration every 10,000–15,000 cycles (sourced: Leeham, Mar 2024)

at 1,000–1,500 cycles a year (sourced: Aircraft Value News), which is 6.7 to 15 years with the central cases, 10,000 at 1,000 to 15,000 at 1,500, at 10 years; an alternative rule of thumb of roughly every eight years (sourced: Aviation Week). Spare ratio: about 10% of installed engines now, from about 15% historically (sourced: MTU AEROREPORT, via Part II's research).

Step 1, the visit rate and the implied interval on each base.

$$\begin{aligned} \text{on } 24,000: & 2,300 \div 24,000 = 9.6\% \text{ a year} \rightarrow 1 \div 0.096 = 10.4 \text{ years between visits} \\ & \text{on } 22,800: 2,300 \div 22,800 = 10.1\% \rightarrow 9.9 \text{ years} \\ & \text{on } 19,000: 2,300 \div 19,000 = 12.1\% \rightarrow 8.3 \text{ years} \\ & \text{on } 14,200: 2,300 \div 14,200 = 16.2\% \rightarrow 6.2 \text{ years} \end{aligned}$$

Step 2, the same at the top of GE's range.

$$2,400 \div 24,000 = 10.0\% \rightarrow 10.0 \text{ years}; \quad 2,400 \div 14,200 = 16.9\% \rightarrow 5.9 \text{ years}$$

Step 3, removing spares from the manufacturer-level base. If about 10% of the 24,000 are spare engines that fly only when installed, the flying base is about 21,600.

$$2,300 \div 21,600 = 10.6\% \rightarrow 9.4 \text{ years}$$

Step 4, comparison with the sourced intervals. An implied interval of about ten years on the 24,000 base matches the mature restoration interval only at the long end of its range, 10,000 cycles at 1,000 a year or 15,000 at 1,500. An interval of 8.3 years on the 19,000 base matches Aviation Week's eight-year rule of thumb. An interval of about six years on the 14,200 base is shorter than any sourced CFM56 interval unless engines fly more than about 1,600 cycles a year ($10,000 \div 6.2 = 1,613$), which is above the sourced utilisation range.

Step 5, the maturity effect. Worked example 3.1 showed about 70% of the fleet with at most one visit in 2024. First runs are longer than mature runs, with the latest variants often flown almost to the LLP limit of 20,000 cycles or more before the first removal (Part I §6.2). A fleet in which most engines are still on a long first run generates fewer visits a year than the mature interval alone would predict, which pushes the implied interval on any base upward. First-run intervals by thrust rating were not retrieved (Part I's unknown U-D1-06).

So the 24,000 base is consistent with the sourced mature interval only if it includes a few thousand spare, stored or green-time engines, or if the long first runs of a still-maturing fleet are depressing the visit rate, or both. The 14,200 base is consistent only with an interval shorter than any the research sourced. The research recorded this and did not resolve it; which base is right is the single most consequential unknown in this Part, because every per-engine figure in Parts IV to IX divides by it. Limits of the example: the visit count and the base come from different publishers and may define "CFM56" differently (all variants or -5B and -7B only); the 10% spare ratio is an industry generality; and the arithmetic assumes every engine is on a regular cycle, which green-time engines with no further visit planned are not. [GE via Aviation Week, 2026; Aviation Business News, Sept 2025; Safran, 13 Feb

2026; Safe Fly Aviation, 2026; Leeham News, 8 Mar 2024; Aircraft Value News, undated; Aviation Week, "Retirement Uptick Will Restock Used Engine Parts"; MTU AEROREPORT; D9 T1; reconciliation R-001, R-002, R-008]

24. The translation table: each driver, the requirement it creates, what it does not stress

Sections 18 to 23 documented the drivers. This section is the device that connects them to the rest of the primer.

TRANSLATION TABLE

The translation table is Part III's organising device: one row per documented demand driver, stating the aftermarket requirement the driver creates and what it does not stress. "Stress" means that the driver raises the quantity or the price of that requirement; the last column records what the driver leaves untouched or pushes the other way. The requirements are the five of aftermarket demand (§18), shop visits, modules, spare engines, used material and leases, with approved alternative parts (PMA, the Parts Manufacturer Approval route, Part VI §37) and capital added because several drivers act on them. Rows are numbered T-1 to T-14 and the numbers are stable; Parts IV to XI open by naming the rows they answer. [D9 §5; outline §24]

MOD-1

Mod-1 is FTAI Power's 25 megawatt (MW) trailer-mounted natural-gas generator set built around a CFM56 core, launched as a platform on 30 December 2025, with first delivery targeted for the fourth quarter of 2026 and 100 units planned for 2027. It is named here because it is a new sink for CFM56 engines outside aviation, which makes it a demand driver in this table. Part IX §69 carries the full treatment. [FTAI release, 30 Dec 2025; D8 §2.3]

How to read the table. Each row names the driver with the section of this Part that documents it and its principal source. The middle column states the requirement in the aftermarket's own units: visits, modules, spare engines, used material, approved alternative parts, leases, capital. The last column is as important as the middle one. A driver that raises shop-visit demand but removes teardown feedstock has done two things, and a reader who carries only the first will misread Part V. Nothing is ranked; the rows are in the order the drivers appear in this Part, with the three drivers documented outside this Part at the end.

EXHIBIT 3.14 — THE TRANSLATION TABLE: FOURTEEN DOCUMENTED DRIVERS, THE AFTERMARKET REQUIREMENT EACH CREATES, AND WHAT EACH DOES NOT STRESS

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-1	Low CFM56 retirements: about 1.5% of the installed base in 2025, about 2% expected for 2026, "below 1%" in 1Q 2026, against a 3–4% plan (§19.3; GE via Aviation Week and Leeham, 2026; GE investor-relations page)	More engines accruing cycles → sustains 2,300–2,400 shop visits a year through 2028; more LLP replacements as retained engines reach cycle limits; more full performance restorations in place of green-time run-outs; module demand for LLP-expired modules	Starves teardown feedstock: fewer used parts and green-time engines, higher used-part prices and spare-engine rents (see T-9); does not stress LEAP or GTF spares; does not change shop capacity
T-2	GTF powder-metal groundings: 835 PW1100G-powered aircraft grounded or stored in late 2025; AOG down more than 20% from the 2025 peak by January 2026 (§20.1; Cirium via trade press; RTX via Leeham)	A320ceo lift to replace grounded neos → ceo lease extensions and short-term leases; higher ceo utilisation → V2500 and CFM56-5B shop visits brought forward; spare V2500 and -5B engines; green-time -5B and V2500 modules for airlines that had planned to retire ceos	Does not create CFM56-7B demand except through general narrowbody scarcity; does not add used-material supply, since it suppresses ceo retirements; creates no CFM56 work on LEAP-type engines
T-3	LEAP durability shortfall and the July 2026 kit: early removals at 2,000–6,000 cycles; LEAP-1B HPT kit certified 18 Jul 2026 with production cutover early 2027; about one-third of the -1A fleet upgraded (§20.2; CFM release; Visual Approach; ePlaneAI)	LEAP shop visits earlier than plan, mostly in the manufacturer's network under long-term agreements; LEAP spare engines; indirectly, slower ceo and 737NG retirements by operators whose neo and MAX fleets are under-available	Creates no CFM56 shop visits, modules or used material; once the kit cuts over, LEAP time on wing lengthens and the indirect support for the old fleet fades

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-4	737 MAX production rate cap: 38 a month from January 2024, 42 from October 2025, 47 reviewed May 2026; 260 MAX delivered in 2024 and 440 in 2025 (§21.1; FAA via trade press; CAPA; Wikipedia compilation)	737NG lease extensions; 737NG utilisation held high → CFM56-7B shop visits and LLP replacements; -7B spare engines; -7B green-time modules for carriers deferring NG retirement	Does not touch A320ceo, -5B or V2500 demand directly; as the cap lifts toward 52, the effect reverses, with more NG retirements, more -7B teardown supply and fewer -7B visits later in the decade; United, WestJet and Copa NG retirements are the counter-case
T-5	Airbus delivery shortfall and the rate-75 deferral: 793 delivered against 820 originally targeted in 2025; gliders 60 → 32 → 0; rate 75 only by end-2027 (§21.2; Air Data News; Leeham)	A320ceo extensions and higher ceo utilisation → -5B and V2500 shop visits; ceo spare engines; keeps ceo part-out low	Does not stress -7B; reduces used-material supply; converts planned airline capital spending on new aircraft into aftermarket operating spending on old ones; reverses as the rate rises
T-6	Lessor extension behaviour: 75–90% of expiries extended; about 12 737NGs and fewer than 25 A320ceos available; ten-year-old 737-800 rents up 44% against January 2022; terms 4–6 years (§22.3; Cirium Ascend; IBA; Air Lease 10-K)	Maintenance events paid by the incumbent airline inside the extended term rather than deferred to redelivery → shop visits and LLP replacements; higher lease rates on mid-life aircraft and engines; capital held in aircraft that stay on lease	Reduces the flow of aircraft into used-sale and part-out markets: fewer acquisition candidates for mid-life buyers and less teardown feedstock; fewer end-of-lease return-condition visits; does not change the total cycles the fleet accrues
T-7	OEM parts-price escalation: new spares about 10% a year in 2022–23 (double-digit on 1 Nov 2022, high-single-digit in Aug 2023); Safran's rule of "above inflation by 3 to 4 points"; the -7B LLP set about 7% a year compounded 2008–2025 (Part I §4; Part IV §28; Aviation Week, late 2023)	Demand for substitutes on the material line, which is 60–70% of every visit: used serviceable material, PMA parts, repair instead of replacement, module exchange instead of overhaul; raises the dollar value of every shop visit whoever performs it	Does not change the number of removals; does not stress labour or slots; does not touch LLPs, for which no PMA exists, so the LLP line escalates unsubstituted

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-8	MRO capacity shortfall and turnaround: slot waits 2–6 months; CFM56 overhaul 90–120 days; demand to exceed capacity by more than 17% by the end of the decade (Bain); two-thirds of operators short of technicians (Oliver Wyman) (§23.4; Part IV §27)	Spare engines and spare-engine leases, since each removal ties up a spare for longer; module exchanges and quick-turn or hospital visits as substitutes for full overhauls; exchange pools and used material to shorten shop time; capital tied up in inventory held for exchange	Does not change the number of removals the fleet generates; does not add teardown supply; shifts where and how the work is done, not whether; GE reports turnaround improving (R-019, R-020)
T-9	Used-material and feedstock scarcity: used-material supply "remains thin" (AVM, 2025); a "hypercompetitive feedstock environment" and whole-asset sales down from 20 engines to 13 (AerSale, 5 Mar 2026); CFM56 values up as much as 50% in two years (ePlaneAI); -7B rents about 34% above 2019 (Safe Fly, consultancy) (§19.4; Part V §33)	Higher prices for used parts and green-time engines; more new OEM parts bought because used ones are unavailable; PMA as the other substitute; capital to acquire the feedstock that does exist, including whole aircraft bought for their engines	Does not add shop capacity; does not change the removal count; is a consequence of T-1, T-4, T-5 and T-6 rather than an independent cause, and reverses when they do
T-10	Data-center power demand and the gas-turbine shortage: global data-center electricity more than doubles to about 945 TWh by 2030 (IEA, Apr 2025); GE Vernova gas-turbine backlog plus reservations 116 GW at Q2 2026 with slots selling for 2031; Siemens Energy lead times of three-plus years (Part IX §71; D8)	A new sink for CFM56 cores outside aviation: 100 Mod-1 units in 2027 would consume about 100 engines a year on a one-core-per-unit reading, competing with teardown and exchange for run-out engines; capital for engine inventory held for conversion	Creates no aviation shop visits; adds no used material, since it removes cores from the pool; on a manufacturer-level base at 1.5–2%, 100 engines are about 21–28% of a year's retirements, or about 35% on the consultancy base of 14,200 at 2%; engines per unit undisclosed

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-11	Freighter conversion: more than 260 737-800 freighters in service and more than 300 in the pipeline (mid-2025); conversion at 21–22 years; IBA warns of "oversupply of narrowbody conversions" (6 Sep 2026) (§19.1; §22.3; Acumen; Cargo Facts; IBA)	Engines with green time or a fresh shop visit for 20-plus-year-old airframes: -7B green-time modules and shop visits; keeps -7B engines on wing past passenger retirement age	Removes airframes and their engines from the teardown pool; does not stress -5B or V2500, since A320ceo conversion is smaller and not quantified here; cargo-market dependent, and the appraiser's oversupply warning cuts the other way
T-12	Higher utilisation of mature narrowbodies: mature passenger-narrowbody utilisation "rising significantly" (§19.2; IBA via AviTrader, Sept 2024)	Faster accumulation of cycles and hours → shop visits and LLP replacements earlier in calendar time; faster exhaust-gas-temperature-margin erosion	Creates no used material; does not change the installed base; ends when replacement aircraft arrive and utilisation normalises
T-13	GTF groundings easing: PW1100 AOG down 25% year to date, maintenance output up 40%, turnaround down 23% (RTX, 27 Jan 2026); GTF Advantage certified 2026 (§20.1; LeeHam; Aviation Today)	Over 2026–28, returning neos reduce the need for ceo lift → ceo extension rates and used ceo lease rates soften; more ceo retirements → more -5B and V2500 teardown supply later	Does not reduce the near-term ceo shop visits already induced; the pace is uncertain, with trade commentary putting full resolution at 2027–28
T-14	CFM56 plateau-then-decline and LEAP aftermarket parity: 2,300–2,400 visits a year through 2028 with the apex at 2027–28 and a gradual fade (GE, 2026); about 2,000 CFM56 visits by 2030 implied (Safran via Aviation Week); LEAP revenue parity in 2028 (Safran Capital Markets Day 2024) (§23.1)	Long run: shop capacity reallocates to the LEAP; teardown supply rises as retirements normalise (McKinsey: about 2.7% of the aircraft fleet a year from 2028)	Does not stress 2026–27 CFM56 demand; the timing depends on the retirement rate, which GE revised three times in one year (R-004)

Source: every figure is cited in the section of this Part named in the row, with the sources listed at the end; rows T-7, T-9 and T-10 draw on Part I §4, Part V §33 and Part IX §71 and their dossiers (D2 §4.4, D3 §2.6, D8 §2.10 and §4.3). Structure from D9 §5, extended with three rows for drivers documented by sibling dossiers. Reconciliation R-001, R-004, R-006, R-007, R-019, R-020, R-028, R-031, R-034.

The paragraphs below explain the mechanism in each row, in the same order.

T-1, low CFM56 retirements. The retirement rate drives both sides of the market at once (§19.4). On the demand side, an engine not retired keeps flying and keeps accruing cycles, so it reaches its next performance restoration and, eventually, its LLP limits. Worked example 3.2 put the gap between the planned 3–4% and the realised 1.5–2% at about 400 engines a year on any manufacturer-level base, and about 1,200 engines over 2024 to 2026. Those engines are why GE and Safran's 2,300–2,400 visits a year hold through 2028 rather than fading from 2026. An engine kept past its planned retirement also changes workscope: an owner who had planned a green-time run-out to retirement now needs a full restoration and a new or used LLP set to run another interval, which is Part IV §30's four-option problem. On the supply side the same engine is one not torn down, so the row's third column hands over to T-9. The row does not touch the new engines' spare pools and does not add a single shop slot.

T-2, GTF powder-metal groundings. Every Pratt & Whitney-powered A320neo on the ground needs its flying replaced, and the practical replacement is an A320ceo on the same type rating (§20.1). The ceo that replaces it flies cycles it would not otherwise have flown. Worked example 3.3 put the ceo-engine visits brought forward at 47 to 250 a year depending on whether the grounded count is RTX's planning average of 350 or Cirium's late-2025 point count of 835. Those visits fall on the V2500 and the CFM56-5B in proportion to the ceo fleet's engine mix, which is unknown (§18.2). The row also raises ceo lease extensions, short-term ceo leases and spare -5B and V2500 engines, and it creates demand for green-time -5B and V2500 modules from airlines that had planned to retire ceos and now must keep them through one more interval. It does not touch the -7B except through general narrowbody scarcity, and because it suppresses ceo retirements it removes ceo teardown feedstock rather than adding it.

T-3, LEAP durability and the July 2026 kit. A LEAP that comes off wing at 4,000 cycles instead of 10,000 generates a LEAP shop visit and needs a LEAP spare, and about half the LEAP fleet is under a long-term agreement, so most of that work flows to the manufacturer's network (§20.2). None of it is CFM56 work. The row's effect on the CFM56 aftermarket is entirely indirect: an operator whose A320neo or 737 MAX fleet is less available than planned keeps its ceo or 737NG flying, which Willis Lease's filing names as a compounding factor. The LEAP-1B kit's certification on 18 July 2026 and its production cutover in early 2027 are the dates at which, on CFM's statement, new MAX engines stop coming off early. From then the indirect support for the 737NG fades, which is why the row's third column carries its own reversal. The LEAP-1A was about a third upgraded at the research date, so the A320ceo side of this row was already fading.

T-4, the 737 MAX production rate cap. Worked example 3.4 showed that the cap alone held back on the order of 170 737NG replacements a year in 2024 and 2025 at a one-for-one

replacement ceiling, or about 77 at Cirium's replacement ratio. Each 737NG not replaced is extended, flown hard and maintained: CFM56-7B shop visits, LLP replacements, spare -7B engines, and green-time -7B modules for carriers deferring retirement. This row is the -7B's counterpart to T-2 and T-5, which are -5B and V2500 rows, and the three must not be fused. The cap is lifting, from 42 toward 47 and 52, and the backlog of more than 4,800 is owed rather than cancelled, so the row reverses: more NG retirements, more -7B teardown supply, and fewer -7B visits later in the decade. United, WestJet and Copa have already begun the retirements that the reversal implies.

T-5, the Airbus delivery shortfall. Airbus delivered 793 aircraft in 2025 against an original target of 820 and now expects A320neo-family rate 75 only by the end of 2027 (§21.2). Each undelivered neo is a ceo its airline cannot release, so the row raises A320ceo extensions, ceo utilisation, -5B and V2500 shop visits and ceo spare engines, and it keeps ceo part-out low. The airline that waits for a late neo still pays for ceo maintenance in the meantime, which is the conversion of planned capital spending into aftermarket operating spending described in §21.3. The row does not stress the -7B. Like T-4 it reverses as the rate rises, because the 8,754-aircraft backlog is a deferral of replacement, not a cancellation.

T-6, lessor extension behaviour. When 75 to 90% of expiries are extended, the maintenance events that would have fallen at a lease transition fall inside the extended term instead, and the incumbent airline pays for them (worked example 3.5). The row therefore raises shop visits and LLP replacements inside extended terms, raises lease rates on mid-life aircraft and engines, and holds lessors' capital in aircraft that stay on lease rather than being sold. Its third column is the one that matters for the mid-life buyer and the teardown trader: only 10 to 25 aircraft of every 100 come back, so the flow of acquisition candidates and teardown feedstock shrinks to that trickle. The row does not change the total number of cycles the fleet flies; it changes who pays for the maintenance those cycles cause and when. The direction of mid-life rents is contested between a new-aircraft lessor's filing and an appraiser's commentary (R-031), and the row carries that tension.

T-7, OEM parts-price escalation. Part I §4 and Part IV §28 own this driver; it is in the table because it acts on every row above it. New spare parts rose about 10% a year in 2022 and 2023, with a double-digit increase on 1 November 2022 and a high-single-digit increase in August 2023, and Safran's stated rule is pricing "above inflation by 3 to 4 points ... at least for some years"; the -7B LLP set compounded about 7% a year from 2008 to 2025 on the figures Part I assembled. [Aviation Week, late 2023; D1 example C] Because 60 to 70% of a shop visit's bill is material priced off the manufacturer's catalogue, escalation flows into every visit whoever performs it (Part I §7). The requirement it creates is for substitutes on the material line: used serviceable material (Part V), approved alternative parts (Part VI), repair instead of replacement, and module

exchange instead of overhaul (Part VII). It does not change how many engines come off wing, it does not stress labour or slots, and it leaves the LLP line unsubstituted, since no PMA exists for a CFM56 life-limited part. [D4 U7]

T-8, MRO capacity shortfall and turnaround. A slot wait of two to six months plus a turnaround of 90 to 120 days means an engine removed today returns in five to ten months, and for that whole period a spare must fly in its place (§23.4). So the first requirement this row creates is spare engines and spare-engine leases, in proportion to the delay rather than to the removal count. The second is substitution of form: a module exchange or a quick-turn visit returns the aircraft to service in weeks rather than months, so the shortage of full-overhaul slots is itself demand for exchanges, exchange pools and the inventory of serviceable modules that an exchange requires, which is capital. The row does not change the number of removals the fleet generates and does not add a single teardown. GE's filing reports turnaround improving and material input up more than 40%, against trade press and a consultancy survey that report the constraint persisting (R-019, R-020); the row carries both.

T-9, used-material and feedstock scarcity. This row is a consequence of T-1, T-4, T-5 and T-6, recorded as its own row because it creates its own requirements and because Part V answers it directly. Fewer retirements mean fewer teardowns; fewer teardowns mean less used material and fewer green-time engines; and the shops that would have used them buy new parts instead, at escalating catalogue prices (T-7). The requirements are higher prices for used parts and green-time engines, more new-parts purchases, approved alternative parts as the other substitute, and capital to acquire what feedstock exists, including whole aircraft bought for their engines. The observable figures are consultancy and trade-press ones and are labelled so: run-out -7Bs bought at \$0.8–1.2M, -7B rents about 34% above 2019, CFM56 values up as much as 50% in two years, and AerSale's whole-asset sales falling from 20 engines to 13. [Safe Fly Aviation, 2026; ePlaneAI; AerSale, 5 Mar 2026] The row reverses when its causes do, which the third columns of T-4, T-5 and T-13 describe.

T-10, data-center power demand and the gas-turbine shortage. Part IX owns this driver. It is in the table because it is the one driver that removes CFM56 engines from aviation for a reason unrelated to aviation. The International Energy Agency (IEA) expects global data-center electricity use to more than double to about 945 terawatt-hours by 2030; GE Vernova's gas-turbine backlog plus reservations reached 116 gigawatts (GW) at the second quarter of 2026 with slots being sold for 2031; Siemens Energy reports lead times of three years and more. [IEA, Apr 2025; Power Engineering, Jul 2026; Oilprice, 2026] Into that shortage FTAI Power sells Mod-1, and 100 units in 2027 would consume about 100 CFM56 engines a year if each unit uses one core, which FTAI has not stated. [D8 §2.10, U3] The requirement created is competition for run-out engines: a core that becomes a generator set is a core that is not torn down for parts and not rebuilt for

exchange. Worked example 3.2 gave the scale on both bases the reconciliation requires: 100 engines are about 21 to 28% of a year's retirements on a manufacturer-level base at 1.5 to 2%, and about 35% on the consultancy base of 14,200 at 2%. [reconciliation C.8] The row creates no aviation shop visit and adds no used material.

T-11, freighter conversion. A conversion candidate is a 20-plus-year-old airframe that needs engines with life on them (§19.1). So conversion creates demand for green-time -7B modules and for shop visits on engines that would otherwise have been retired with their passenger airframes, and it keeps those engines on wing past passenger retirement age. Its third column is the teardown trader's: each converted 737-800 is an airframe and two engines removed from the part-out pool. The row is specific to the -7B, because A320ceo conversion is a smaller programme that the research did not quantify, and it is contested at the research date, with more than 260 737-800 freighters in service and more than 300 in the pipeline on one side and IBA's September 2026 warning of oversupply on the other (R-034).

T-12, higher utilisation of mature narrowbodies. IBA observed in September 2024 that delivery delays had led operators to maximise daily utilisation across all age groups, with mature passenger-narrowbody utilisation "rising significantly". [IBA via AviTrader, 24 Sep 2024] Utilisation here means flying intensity, cycles and hours per aircraft per period, not the lessor's on-lease share that Part VIII §58 uses; the registry records three meanings of the word and this row uses only the first. [registry C-3] An engine flown 1,500 cycles a year reaches a 10,000-cycle interval in under seven years; at 1,000 cycles a year it takes ten. The row therefore brings shop visits and LLP replacements forward in calendar time and erodes exhaust-gas-temperature margin faster, without adding a single engine to the base or a single teardown to supply. It ends when replacement aircraft arrive and utilisation normalises.

T-13, GTF groundings easing. This is the reversal of T-2, recorded separately because it was already under way at the research date. RTX reported PW1100 AOG down 25% year to date, maintenance output up 40% and turnaround down 23% in January 2026, with the GTF Advantage certified for entry into service later in 2026 (§20.1). Each neo that returns to service releases the ceo that was flying in its place. Over 2026 to 2028 that softens ceo extension rates and used ceo lease rates and allows the ceo retirements that T-2 and T-5 suppressed, which in turn adds -5B and V2500 teardown supply later in the decade. The row does not reduce the ceo shop visits already induced, since an engine restored in 2025 has been restored, and its pace is uncertain, with trade commentary putting full resolution at 2027 to 2028.

T-14, the CFM56 plateau-then-decline and LEAP parity. GE's 2026 curve holds 2,300 to 2,400 CFM56 visits a year through 2028 with the apex around 2027 to 2028 and a gradual fade after; Safran's earlier framing implied about 2,000 CFM56 visits in 2030 and LEAP aftermarket revenue at parity with the CFM56 by 2028 (§23.1). The long-run requirement is a reallocation:

shop capacity moves to the LEAP, and teardown supply rises as retirements return toward McKinsey's about 2.7% of the aircraft fleet a year from 2028. The row does not stress 2026 to 2027 CFM56 demand. Its timing is the least certain figure in the table, because it depends on the retirement rate, which GE revised three times in one year, and because the two CFM partners' curves disagree on level and shape (R-004, R-006, R-007).

EXHIBIT 3.15 — WHERE EACH AFTERMARKET REQUIREMENT IN THE TABLE IS TREATED IN THE PRIMER

Requirement named in the middle column	Owner Part and sections	Rows in which it appears
Shop visits, worksopes and the shop-visit market	Part IV §25–§30	T-1, T-2, T-4, T-5, T-6, T-8, T-11, T-12, T-14
Used serviceable material, teardown, feedstock and green-time engines	Part V §31–§36	T-1, T-2, T-4, T-5, T-6, T-7, T-9, T-10, T-11, T-13, T-14
Approved alternative parts (PMA) and repairs	Part VI §37–§44	T-7, T-9
Modules, module exchange and the Module Factory	Part VII §45–§57	T-1, T-2, T-4, T-7, T-8
Leases of aircraft and engines, extensions, spare engines and the capital behind them	Part II §12–§17; Part VIII §58–§67	T-2, T-4, T-5, T-6, T-8, T-9, T-13
CFM56 cores as power-generation feedstock	Part IX §68–§74	T-10
Capital: inventory, acquisition, third-party vehicles	Part VIII §61–§66; Part X §81–§82; Part XI §86	T-6, T-8, T-9, T-10

Source: outline.md section numbers; the rows of Exhibit 3.14. Compiled for this Part. The mapping says where each requirement is treated; it does not say which rows each Part will name as answered, which each Part states for itself.

The table closes the first-principles build. Everything a reader needs to interpret a quantity in Parts IV to XI is now in place: how the engine works and what it costs (Part I), how money moves around it (Part II), and which of fourteen documented drivers is pushing which requirement in which direction (this Part). Where two sources disagree on a driver's size, the table carries the tension in its row and the section behind it carries both figures. Where the research could not find a figure, the unknowns list below says so.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-001	CFM56 in service: ~23,000 (CFM, undated; GE 2019 / Safran 2024 via D2); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); "over 20,000" -5B/-7B (Chromalloy, Oct 2025); >19,000 and ~14,200 (Safe Fly, consultancy blog)	§18.1, Exhibit 3.1; worked examples 3.1, 3.2, 3.6
R-002	Safe Fly's ~14,200 read as the -7B fleet (D3) or all CFM56, ≈6,200 -5B and ≈7,500 -7B (D8)	§18.1, §18.2, Exhibit 3.1
R-003	-7B: "in excess of 8,000" installed (CFM product page, undated) against >15,000 delivered and >7,000 aircraft at two each	§18.2
R-069	Deliveries >35,000 (CFM; trade press) against "well over 30,000" (unsourced)	Exhibit 3.1
R-010	V2500 in service: 5,286 (Aviation Week, 2023) against ">13,000" (low-grade vendor page, mid-2026)	§18.2, Exhibit 3.2
R-012	Fleet maturity: 57% / 21% (GE, Jun 2019); ~70% zero or one (Safran, 2024); "over half" / "another quarter" (Safe Fly, 2026)	§18.3, Exhibit 3.3; worked example 3.1
R-004 (D9 T3)	GE retirement rate: 3–4% → 2–3% → 1.5–2% (CFO via Aviation Week, 2026); 2025 ~1.5%, 2026 ~2% (Leeham, 22 Jan 2026); 1Q 2026 "below 1%" (GE investor-relations page)	§19.3, Exhibit 3.6; worked example 3.2
R-005 (D9 T2)	Aircraft retirements 2024: fewer than 200 (IBA, Sept 2024) against 587 (Cirium via Informa); derived ~410 (McKinsey) and ~1,130 (Cirium five-year projection); IBA's 2020 peak against Cirium's 2019 > 2020	§19.2, Exhibit 3.5
D9 T6, T7	GTF grounded aircraft: ~350 average 2024–26 (RTX planning, 2023) against 835 grounded or stored (Cirium, late 2025); "720 of 1,912" with undefined denominator; RTX's "down >20%" without a level	§20.1, Exhibit 3.7; worked example 3.3
R-006	CFM56 shop visits: 2,300–2,400 a year 2026–28 (GE, 2026) against ~2,500 peak 2025–26 (Safran, c. 2023/24); IBA "+40%" 2024 → 25 and Safran "mid-single-digit" 2025 unreconciled	§20.2, §23.1, Exhibit 3.12
R-007 (D9 T4)	Shape: peak 2025–26 (Safran) against plateau to 2028 with apex 2027–28 (GE) against first-visit peak 2027 (Visual Approach) and 737-700 decline after 2027 (Acumen)	§20.2, §23.1, Exhibit 3.12
R-008 (D9 T1)	Interval: 10,000–15,000 cycles (Leeham) against ~8 years (Aviation Week) against ~10 years implied by 2,300 visits on 24,000 engines	§23.4, worked example 3.6

Id	The two (or more) figures	Where in this Part
R-009	V2500 shop visits 2025: >800 (RTX CFO via Aviation Week, 2026) against ~1,400 forecast (Aviation Week data tool)	§23.2, Exhibit 3.13
R-011	Removal driver: EGT margin "most influential" (Aircraft Commerce, 2003/2008) against LLP-driven for the latest -5B/-7B (Aircraft Commerce Issue 50; FlightGlobal); D9 lists three clocks unranked	§22.1
R-019, R-020	Turnaround and parts: 90–120 days and slot waits 2–6 months (trade press; Bain) against ~90 days sustaining and material input +40% (GE 10-K FY2025; GE, 27 May 2026); Oliver Wyman labour survey	§23.4
R-028 (B-5)	-7B engine rent: ~\$100,000/month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time); both about a third above 2019	§22.3, Exhibit 3.11
R-031	Mid-life narrowbody rents: rising (Air Lease 10-K FY2025; Cirium +44%) against 12-year 737-800 ~\$228,000 (–11%) and A320-200 ~\$220,000 (–13%) (IBA, Jul 2025) and IBA's Sept 2026 warning	§22.3, Exhibit 3.11
R-034	Conversion demand: FTAI–AEI collaboration (7 Jul 2026), >260 737-800Fs and >300 pipeline against IBA's "oversupply of narrowbody conversions" (6 Sep 2026)	§22.3; row T-11
D9 T5	CFM's "in excess of 8,000" -7B installed against >14,000 implied by >7,000 aircraft	§18.2
D9 T9	Airbus 2025 target: 820 (Argus) against ~790 revised and 793 delivered	§21.2, Exhibit 3.10
D9 T10	"Sevenfold" shop-visit rise by 2040 (IATA via AFM) against Safran's LEAP curve and the >4,000 CFM-stable figure	§23.3

Unknowns carried in this Part

- **U-D9-01 (D9) — U1 — 2026 in-service counts and average ages for the A320ceo family and the 737NG, separately, with parked share.** Only a combined A320 family in-service figure (11,308, July 2026 [S13]) and a 2023 stored-A320ceo figure (~570, avg 17.1 years [S14]) surfaced. Resolve: Cirium Fleets Analyzer or IBA Insight extracts, or the fleet tables in Aviation Week's 2026 Fleet & MRO Forecast.
- **U-D9-02 (D9) — U2 — Date of the "47% of 737NGs / 45% of A320ceos aged ≤10 years" statistic [S77/S78].** Given production ended in 2019/2022, the proportion in 2026 must be lower; the figure is probably from 2021-22. Resolve: read the ISTAT article.

- **U-D9-03 (D9) — U3 — Definition of the ~24,000 CFM56 "in service" figure [S78]** (installed only, or installed plus spares plus stored) and its split between -5B, -7B and the older -3/-5A/-5C variants. Resolve: GE Aerospace or CFM investor materials with a fleet breakdown; the GE Bernstein presentation (27 May 2026) [S6] may contain it but its content did not surface.
- **U-D9-04 (D9) — U4 — V2500 engines in service, 2025-26, from IAE/RTX or Cirium.** Only a low-grade figure surfaced (T8). Resolve: IAE/Pratt & Whitney V2500 programme page or RTX investor day fleet slide.
- **U-D9-05 (D9) — U5 — A year-by-year series of GTF AOG counts (average and peak) for 2024, 2025 and 2026 from RTX's own disclosures.** RTX's statements surfaced as percentages (-20%, -25%) rather than levels [S31]. Resolve: RTX Q4 2024, Q4 2025 and Q2 2026 earnings slides/transcripts [S32].
- **U-D9-06 (D9) — U6 — Historical retirement rates for the A320ceo and 737NG specifically (annual counts 2015-2025).** Only aggregate all-type retirements and the CFM56 percentage surfaced. The sentence "retirements in relation to the share of the active fleet as of 2023 was 31% for the 737 and 26% for the A320" appeared in a result [S59/S60 context] but its meaning (share of total retirements by type?) is not clear. Resolve: Cirium or IBA type-level retirement tables.
- **U-D9-07 (D9) — U7 — The 2026 Aviation Week Fleet & MRO Forecast's type-level shop-visit curve for CFM56 through 2030, and Oliver Wyman's narrowbody-engine subtotal by year.** Only headline totals surfaced [S47, S51]. Resolve: the forecast documents themselves.
- **U-D9-08 (D9) — U8 — Labour figures (engine MRO technician headcount, shortfall, wage inflation).** Oliver Wyman cites skilled-worker retirements qualitatively [S47]. D2 may hold these. (Partly supplied: Oliver Wyman, Apr 2026, two-thirds of operators short of technicians, labour-rate inflation 5.5–6.0%; reconciliation C.9.)
- **U-D9-09 (D9) — U9 — Airline-level fleet plans with CFM56 retirement dates.** Spirit's 2023 fleet plan exhibit surfaced as a URL [SEC fleet plan, Aug 2023] but was not read; the FY2025 10-Ks of Southwest, JetBlue and Frontier surfaced but their fleet-plan tables were not extracted [S73-S75]. Resolve: read the fleet-plan sections.
- **U-D9-10 (D9) — U10 — Boeing/Airbus 20-year market outlook replacement-demand figures for single-aisles (the share of deliveries that replace retirements).** Cirium's 2024 fleet forecast (44,500 deliveries, \$2.7tn, 20 years) surfaced [S83] but without the replacement share. Resolve: Boeing CMO and Airbus GMF 2025/2026 editions.
- **U-D9-11 (D9) — U11 — Average AOG duration (wing-to-wing days) for a GTF powder-metal shop visit, by year.** RTX gave a 23% reduction in turnaround [S31] without the base. Resolve: RTX transcripts.

- **U-D2-02 (D2) — U2. Shop-visit counts 2020–2023.** Only the 2019 level (~2,000+) and the 2024 forecast ("topping 2,000") were found. Resolve from Safran capital-markets-day decks (the safran-group.com media downloads 397180 / 448184 returned in search but not read) or the GE Aerospace Bernstein presentation of 27 May 2026.
- **U-D2-10 (D2) — U10. Dates.** Several Aviation Week articles were returned without publication dates (noted "c. 2023/24" or "2025"). The writer should confirm before quoting.
- **U-D3-16 (D3) — U16 —** Actual retirement counts for 2023, 2024 and 2025 from Cirium (only 2019–2022 counts and the 2024–26 shortfall percentage were captured).
- **U-D1-12 (D1) — U12** Current LEAP delivery total and the dates/years of -5A and -5C production end.
- **U-D11-16 (D11) — U16 — GE CFM56 shop-visit count, spare-parts price increases, and any "material solutions" or used-serviceable-material programme wording.** Not returned from [S14]. Resolve: read the CES segment discussion in [S14] and the Q4 2025 and Q2 2026 earnings materials.

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